

Lead Halide Perovskite Quantum Dots for Photovoltaics and Photocatalysis: A Review

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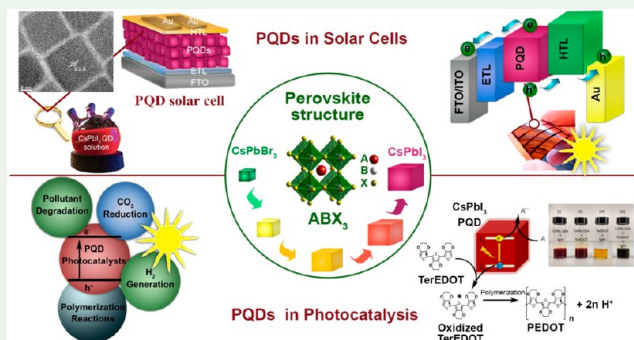
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ABSTRACT: Lead halide-based perovskite quantum dots (PQDs) have recently emerged as an important class of nanocrystal (NC) materials for optoelectronic and photoelectrochemical applications. Thanks to their intriguing features including tunable band gap, narrow emission, high charge carrier mobility, remarkable light-absorbing factors, and long charge diffusion length, there has been a surge in research on lead halide-based PQDs and their applications. In this review, we showcase the fundamentals of PQDs and two principal applications including PQD solar cells (PQDSCs) and photocatalytic conversion. First, a thorough discussion on PQDSCs, their structure, surface treatment, and interface engineering along with their recent progress are presented. It is highlighted that the improvement of the efficiency of PQDSCs from below 10% to beyond 16% in a matter of a few years has turned them into promising candidates for future SC applications. Subsequently, the application of PQDs in photocatalytic reactions such as hydrogen production, CO₂ reduction, and organic compounds' degradation is summarized. Not to mention that, despite the remarkable properties of PQDs in SCs and photocatalysis, the inferior stability of PV devices based thereon under operation as well as their poor tolerance under air, water, light, and heat impede their widespread application. For this, the practical efforts and possible solutions are extensively addressed. Finally, an outlook is provided, addressing further merits, and demerits of each application as well as prospective opportunities.

KEYWORDS: Lead halide perovskite quantum dots, Solar cells, Photocatalysis, Efficiency, Stability



1. INTRODUCTION

Perovskite quantum dots (PQDs) or nanocrystals (NCs) stand out in terms of rigorous research in the span of the last few years by virtue of their merits including inexpensiveness, tunable bandgap, flexible compositional control, multiple exciton effects, and solution processing as well as electronic and surface chemistry properties.^{1–7} QDs display a quantum confinement effect with varied device structures such as bulk heterojunction (BHJ), not readily obtained by thin-film solar cells (SCs).^{1,8–10} Lead halide PQDs (LHPQDs) own special light-emitting traits and have compatibility with industrialized printing technology, which exhibit a promising prospect in next-generation flexible and high-definition displays.¹¹ In addition, colloidal HPQDs on account of their long lifetime, outstanding conductivity, and diffusion distance have been identified as superb photodetectors.¹² What is more, inorganic halide perovskite quantum dots (IHPQDs) materials with relatively high stability,¹³ ultrahigh photoluminescence quantum yield (PLQY), and widely absorption spectra covering the entire visible range¹⁴ have recently been used in photoelectrochemistry. PQDs' unique properties combined with

solution-processable device fabrication enable novel forms of advanced optoelectronic devices for energy harvesting, light generation, photon detection, and photoelectrochemical reactions.

Application-wise and for the sake of energy harvesting, herein, the two intriguing cases are addressed. First, PQDs offer a promising solution for SCs. Their spectral absorption window covering the UV–visible (UV–vis) range up to the near-infrared range combined with short-wavelength infrared (SWIR) range allowed for a jump in power conversion efficiency (PCE) levels from 3.8%^{5,15,16} to beyond 16.6%.¹⁷ Second, in the interest of photoelectrochemical (PEC) systems, the following features could be achieved: (i) wide excitation with which IHPQDs can be excited by a single light

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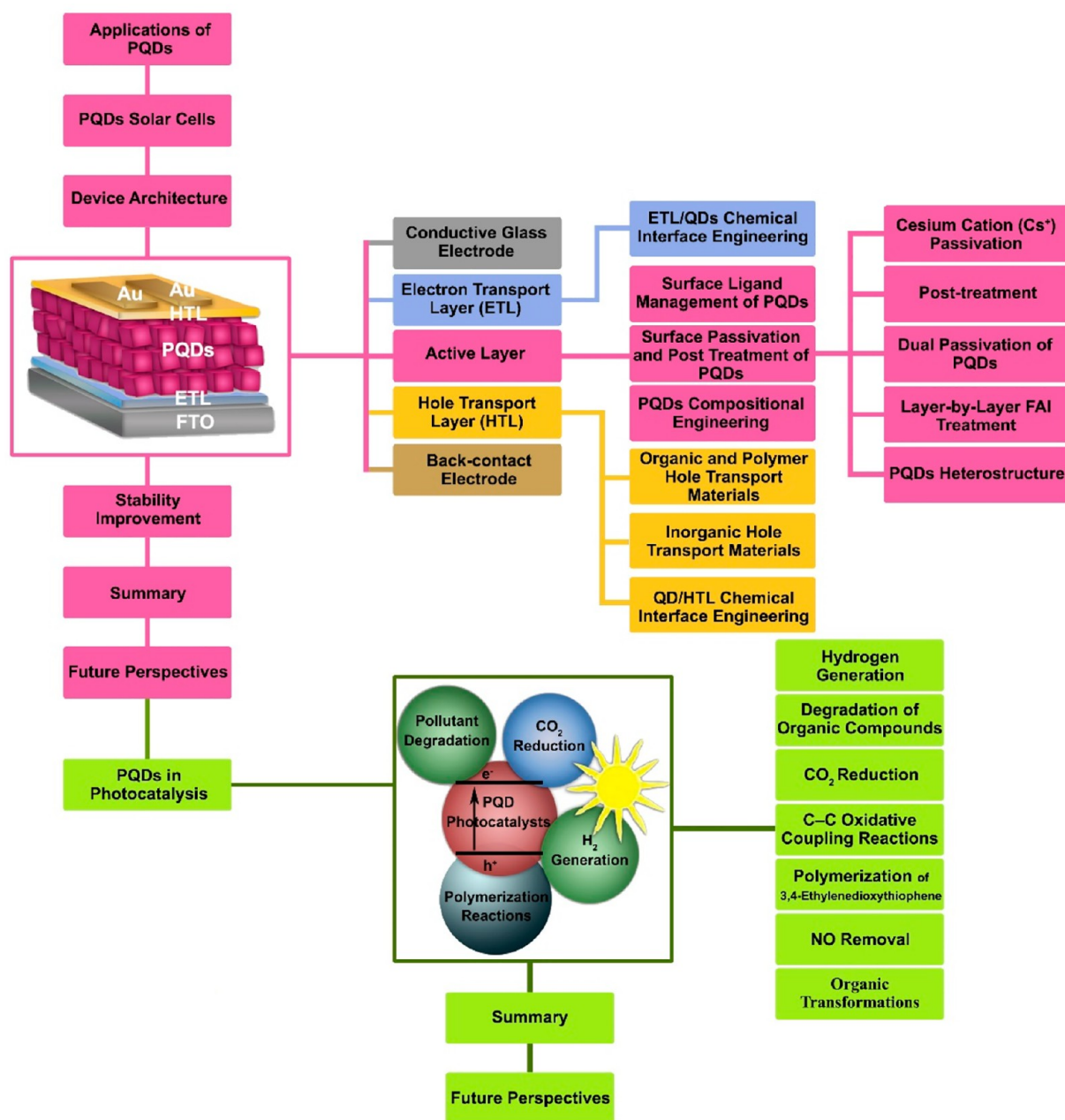


Figure 1. Overall framework of the review.

source in the range of 350–400 nm, and (ii) narrow full width at half-maximum (fwhm) of the emission peak (only 12–42 nm) which is much narrower than that of traditional quantum dots and organic dyes, and its color gamut range can reach 140% of the National Television System Committee (NTCS) standard.¹⁸ Hence, in the last years, PQDs appeared as game-changing nanomaterials for a wide range of optoelectronic devices.

In this review, we thoroughly explore the fundamentals of PQDs in two prominent applications (Figure 1). To begin with, the relentless search for sustainable forms of energy is a pressing solution to the severe energy crunch and a strategy to confront the brutal environmental concerns.¹⁹ Over the last few years, the photovoltaic (PV) community has fiddled with the idea of the fabrication of PVs with halide perovskite solar cells (PSCs) thanks to their high-power conversion efficiency

(PCE).²⁰ Lately, nanometer-sized PQDs have emanated with striking benefits as opposed to their large grain size perovskite thin-film analogues.^{21–25} Thanks to various advantages, QDs are able to exceed the Shockley–Queisser limit²⁶ and further enhance the efficiency of tandem devices. For this, we provide a detailed description of the architecture of PQDSCs composed of multiple layers accompanying the function of each layer.

The story of PQDs within the scope of this review includes their application in photocatalysis as well to contribute to the solution of energy and environment-related issues. Despite the tremendous success of organometallic halide perovskites (OHPs) in optoelectronic applications,^{27,28} they fail to fulfill the criteria in other applications like photocatalysis mainly due to their instability.²⁹ That is why IHPQDs have received significant attention in the field of photoelectrochemistry. For

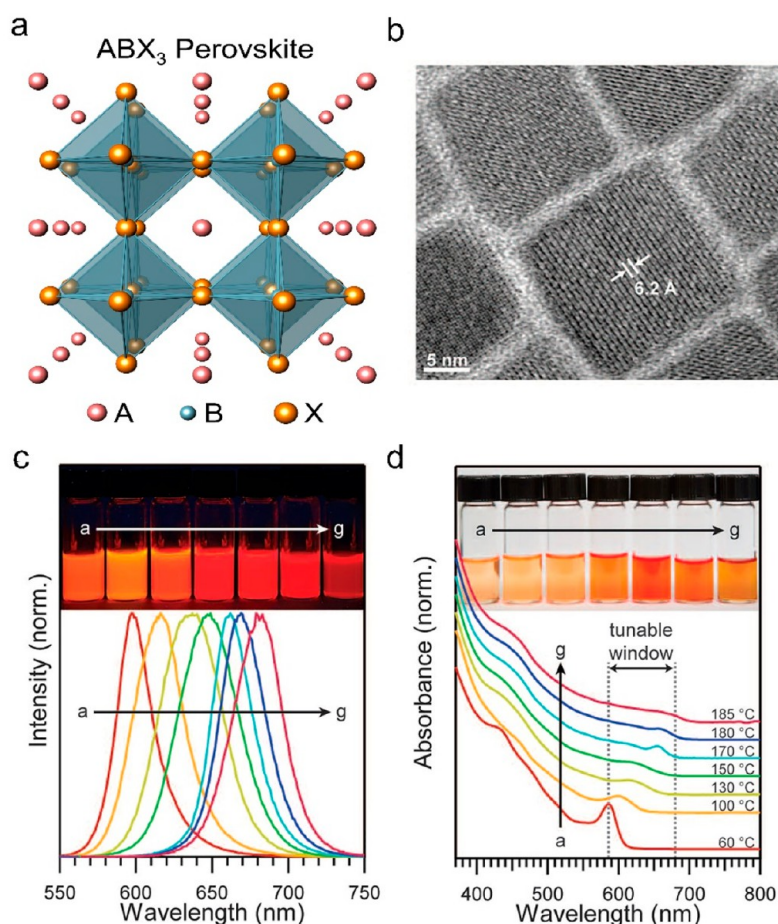


Figure 2. (a) Crystal structure of ABX_3 perovskites QDs. (b) High-resolution TEM of $CsPbI_3$ QDs synthesized at 180 °C. (c) Normalized UV-vis absorption spectra and photographs of $CsPbI_3$ QDs synthesized at (a) 60 °C (3.4 nm), (b) 100 °C (4.5 nm), (c) 130 °C (5 nm), (d) 150 °C (6.8 nm), (e) 170 °C (8 nm), (f) 180 °C (9 nm), and (g) 185 °C (12.5 nm). The numbers in parentheses are the average size obtained from TEM. (d) Normalized photoluminescence spectra and photographs under UV illumination of the QDs from panel c. Reproduced with permission.⁴⁶ Copyright 2016, Science.

this reason, here we have compiled various applications of PQD-based materials in photoelectrochemical reactions differing from photocatalytic hydrogen generation, degradation of organic compounds and antibiotic residues in organic systems, CO_2 reduction, stereoselective C–C oxidative coupling, polymerization reactions, and NO removal to organic transformations.

In essence, herein, we discuss compendious aspects of PQDs including their fundamentals and two pioneer applications.

2. FUNDAMENTALS OF PEROVSKITES: CRYSTAL STRUCTURES, DIFFERENT PHASES, AND BANDGAP ENGINEERING

Perovskites are a large family of ceramic crystalline materials having three-dimensional (3D) structures similar to natural mineral calcium titanium oxide.³⁰ They are the main members of the perovskites NCs (PNCs), which can be described by the general chemical formula of ABX_3 (see Figure 2a), in which B cations are 6-fold coordinated to X anions creating corner-sharing BX_6 octahedra and A cations fill the cubo-octahedral holes for maintaining charge neutrality of the system.^{31,32} For the case of metal halide perovskite, B can be a divalent metal such as Pb^{2+} , Sn^{2+} , Cu^{2+} , Ni^{2+} , Co^{2+} , or Mn^{2+} and X a halogen (I^- , Br^- , or Cl^-). Generally, “A” cations such as Cs^+ ,

methylammonium (MA^+), or formamidinium (FA^+) form a 3D framework with the octahedral network. Unlike more covalent semiconductor NCs (e.g., $CdSe$,³³ InP ,³⁴ $AlSb$ ³⁵), fully crystalline metal halide perovskites have ionic bonding, which affects their environmental stability, electronic structure and defect tolerance, and ligand binding and fluxionality.^{36–38} The low stability and highly ionic nature of 3D perovskites urge researchers to work on the low dimensional perovskites. For that, 2D LHPs having a tetragonal phase of Cs^+ and $[Pb_2X_5]^-$ polyhedron layers were explored.^{37,39} Their structure can be considered as a 3D ABX_3 perovskite cut into one repeat unit thick slice along the $\langle 100 \rangle$ direction, where A and X ions within the slice planes are cut into halves.^{32,40} A_2PbX_4 perovskites are another type of 2D perovskites consisting of corner-sharing $[PbX_6]^{4-}$ octahedra and bulky cations layers.^{37,41} On the other hand, 0D perovskites (Cs_4PbX_6) were studied recently,^{42,43} where the PbX_6^{4-} octahedra are individually disconnected and are completely decoupled in all dimensions, and the halide ions are no longer shared between them.^{37,44,45}

Even though thin film or bulk perovskite materials possess higher power efficiency, they have inferior phase stability at room temperature compared to PQDs. Their sufficient stability can be ascribed to nanocrystal surfaces and electronic coupling between the QDs. As can be seen from Figure 2b, the

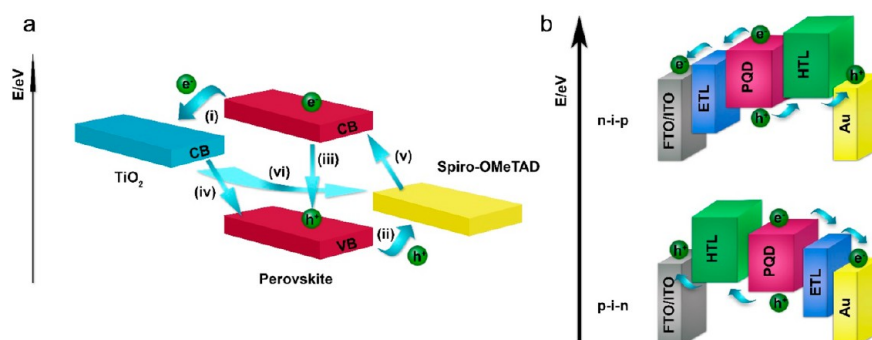


Figure 3. (a) Schematic diagram of energy levels and transport processes of electrons and holes in an HTM/perovskite/TiO₂ cell. (b) Schematic structure and energetics of different configurations of PQDSCs.

transmission electron microscopy (TEM) micrograph of CsPbI₃ QDs reveals an interplanar distance of 0.62 nm, which is consistent with the (100) plane of cubic phase stable at room temperature.⁴⁶ Besides distinctive structural properties, the quantum confinement effect of metal halide perovskites has received significant attention for the facile tuning of bandgap.^{37,47,48}

Since metal halide perovskites possess a soft ionic framework, they exhibit significant polymorphism, manifested by the presence of several different phases within one ABX₃ composition. Therefore, the phase transition between high-temperature low-bandgap (1.5 eV) and low-temperature high-bandgap (>2.8 eV) hexagonal perovskites has been well documented for widely used FAPbI₃ and CsPbI₃ compositions. Density functional theory (DFT) calculations revealed that bandgap was affected by two main factors, including the ratio of cubic (c) and hexagonal (h) close-packed layers and the thickness of blocks of cubic layers. It was reported that there is a nonlinear relationship between the band gap and the ratio of cubic and hexagonal layers in these structures. Since crystallization of perovskites is often complicated, and the formation of various phases with the same stoichiometry is inevitable, the presence of such polytypes in the perovskite matrix is responsible for a decrease in the mobility of charge carriers.⁴⁹

The change of the divalent metal and anionic components change the X–B–X bond impact and tune the bandgap of a metal halide perovskite.^{32,50} By substitution of larger cations such as FA and MA, the lattice of metal halide perovskite expands, and the bandgap decreases.^{32,51} Moreover, by changing the halide from iodide to bromide and chloride, the bandgap increases, and thus emission wavelength produces a blueshift. The composition of divalent metal can tune not only the bandgap but also the absolute energy levels of the valence and conduction bands.^{52,53} Meanwhile, the band gap of QDs can be tunable via size control due to quantum confinement (Figure 2c,d).⁴⁶ At the moment, a large number of perovskites have been identified with a broad spectrum, covering from conductors to insulators.^{54,55} So far, the most intensively investigated colloidal QDs (CODs) have been manufactured from chalcogenides within the zinc blend or wurtzite crystal structure, whereas their perovskite counterparts outperform them in the context of superconductivity, ferroelectricity, colossal magnetoresistance, and so forth.⁵⁶

3. APPLICATIONS OF PEROVSKITE QUANTUM DOTS

3.1. Perovskite Quantum Dot Solar Cells. Despite considerable attention swayed to dye-sensitized solar cells (DSSCs) because of their facile synthesis and low price,^{57–59} the thick absorbing layer (>10 μm) and the discoloration of organic dyes induced by light are still their major demerits. The controversial bottlenecks have convinced researchers to seek another candidate composed of all-solid dyes. This is where PQDs have come under the spotlight. Primarily, perovskite materials were exploited solely as a dye alternative material. Later, the substitution of liquid electrolyte with a solid-state hole conductor acted as a game changer.^{60,61}

The intriguing optoelectronic characteristics of perovskites including broadband light absorption and high absorption coefficient in the visible range (~10⁵ cm⁻¹), to name but two, have made them simultaneously employed as an electron/hole transport layer and light-absorbing layer. It should be noted that a significant reduction of the desired thickness as well as the elimination of barriers to harvesting photogenerated carriers can be achieved as a result of these materials' strong optical absorption. In 2009 Kojima and co-workers⁶² realized that metal halide perovskites play the role of dyes in the harvesting of sunlight. Following this, employing the organic metal halide perovskite (MHP) as a sensitizer under a liquid electrolyte in DSSC delivered a PCE of 3.8%.

Compared to well-known N719 dye sensitizers, perovskites exhibited improved absorption; however, their fragile tolerance in polar solutions resulted in their instant deterioration.⁶³ In 2012, solid-state-based PSCs exhibited an over 2.5-fold improvement in terms of PCE in the absence of liquid electrolytes and reached 9.7%.⁶⁰ Over the last few years, we have witnessed the rapid development of PSCs by virtue of their inexpensive manufacturing process along with high efficiency. In this regard, the highest PCE of 25.5% was obtained for organic–inorganic lead halide PSC in 2020, showing substantial durability⁶⁴ and approaching the 27.6% efficiency for commercially single crystalline silicon SCs.⁶⁵ Nonetheless, large-scale implementation of PSCs appears to be far-fetched due to their insufficient durability when they undergo treatment in different conditions.⁶⁶ To enhance the performance of PSCs and strengthen their durability, different perovskite compounds such as QDs have been manipulated. The objective of this section is to introduce multiple layers of PSC, followed by a detailed description of the function of each layer. Subsequently, a new generation of active layers as QDs for a typical PSC is presented. Eventually, the latest developments of PQDSCs are delineated.

A representative PSC is made up of multiple layers comprising a conductive glass electrode, an electron transport layer (ETL), an active layer (perovskite layer), a hole transport layer (HTL), and a back-contact electrode layer. Once the PSC receives the sunlight, the perovskite layer absorbs photons, generating electrons and holes so-called excitons. Smaller binding energies of excitons in perovskites arising from the large dielectric constant can lead to free electrons and holes, ending up either as current or recombination of carriers to form excitons.⁶⁷ The long lifespan and diffusion length of perovskites come as no surprise, since the mobility of charge carriers is quite high, while recombination rates remain low. Both extended lifespan and long diffusion length are prime reasons behind the remarkable performance of PSCs.⁶⁸ Once the electrons and holes are free, they are assembled by respective ETLs and HTLs. Thereafter electrons are delivered to TiO_2 as the electron-transporting material (ETM) layer and assembled by fluorine-doped tin oxide (FTO) electrode. Holes, on the other hand, are delivered to the hole-transporting material (HTM) layer and assembled by the counter electrode. The photocurrent is produced throughout the external circuit via the connection between the counter electrode and FTO.

Figure 3a depicts the arrangement of energy levels for the transport process of carriers within an HTM/perovskite/ TiO_2 cell. The beneficial processes shown in Figure 3a encompass excitation induced by light in perovskite (process (i)) followed by electron injection to titania (process (ii)). It should be pointed out that electron and hole splitting takes place at the interfacial segment of TiO_2 /perovskite and Spiro-OMeTAD/perovskite accompanied by electron delivery to TiO_2 (process (i)) and hole delivery to HTM (process (ii)) with which charge transport process is accomplished.⁶⁹ Besides the mentioned events, unwanted processes namely recombination of photoexcited carriers, back charge transfer at the interfacial segments of TiO_2 and the HTM with the perovskite (photoluminescence), a backward motion of carriers (process (iv) and (v)), and electrons and holes recombination at the interfacial junction of TiO_2 and the HTM (process (vi)) occur concurrently within the cell, undermining its performance.

3.1.1. Device Architecture. Manufacturing PSCs can be achieved either through a normal arrangement or inverted configuration based on the carrier assemblage electrode, which can be further modulated by employing multiple layers. The structure of PSCs is defined depending on the position of the conductive substrate—FTO or indium tin oxide (ITO)—and deposited layer. For this, if the conductive substrate is deposited on the ETL; it is a normal arrangement or n-i-p configuration. On the other hand, if an HTL is placed on a conductive substrate, it is called an inverse arrangement or p-i-n configuration. Figure 3b illustrates both configurations.^{70,71} PSCs with the configuration of n-i-p can be classified into two planar or mesoporous structures with reference to their ETL construction. The n-i-p mesoporous structures are the original framework of the perovskite PV devices that are currently in common use. Keeping this in mind, the mesoporous layer can intensify charge assemblage through lowering the distance of charge transit, preclusion of current leaking at interfacial segments, and enhancing of absorption of photons because of the light scattering phenomenon. In the early stages, people postulated that mesoporous layers of ETM with large areas affect positively the performance of perovskite devices, considering that electron removal at the interfaces of ETM is

more cumbersome compared to hole removal at the HTM.⁷² That being said, careful control of the fabrication of perovskites and their interfaces, formation of carrier layers, as well as electrodes result in high-efficiency devices in the absence of a mesoporous layer.⁷³

At the same time, for the fabrication of the p-i-n type perovskite device, the following procedure can be adopted. First, a p-type conductive polymer is grown on an ITO-coated substrate. Following the deposition of a thin layer of perovskite, a hole-blocking layer and a cathode either Au or Al are placed on one another, respectively. For the initial constructions, the materials employed as donor and acceptor were perovskites and fullerene, separately.⁷⁴ During the recent rapid development of PV devices, the organic acceptor has been replaced by a thin layer of ETM, resulting in planar perovskites in between two organic charges transiting layers.⁷⁵ For the additional advances in this field, one can mention the switching in selective contact from organic to inorganic compounds. Continuing on this line, the recent usage of NiO and ZnO/TiO_2 layers as respective hole and electron selective contacts has turned perovskites into superior materials as opposed to their analogues.^{76,77} Moreover, manufacturing the mesoscopic p-i-n configuration is subject to the exploitation of HTMs such as $\text{NiO}/\text{mp-Al}_2\text{O}_3$ or $\text{c-NiO}/\text{mp-NiO}$.^{70,71}

3.1.1.1. Conductive Glass Electrode. A conducting glass electrode acts as both a current collector and a substrate to grow multiple layers on it. The transparency of the glass electrode plays a pivotal role in letting sunlight pass through and reach the active region of the cell. Sheet resistance, on the contrary, is in inverse relation to the carrier assemblage of the cell: the lower the resistance, the higher the charge collection. To this end, FTO, with regard to its low sheet resistance and thermal stability is the first choice to be used as a conducting electrode. Sima et al.⁷⁸ proved that contrary to FTO, ITO is not thermally stable, showing a resistance increase from 18 to $52 \Omega/\text{cm}^2$ at a high-temperature heat treatment. Hence, ITO is applicable for the low-temperature fabrication of cells. Moreover, Xu et al. and Hu et al.^{3,6} reported the application of SnO_2 at low temperatures of $120\text{--}150^\circ\text{C}$ as ETL on ITO to construct the SCs. On the flip side, in connection with the application of TiO_2 as ETL at higher temperatures, FTO is preferred.^{46,79} In addition, polymers are alternated by dint of their versatility and affordability. Despite these advantages, they are restrained by the applicable temperatures. For flexible SCs, the conductive electrode should possess several major exclusivities including good endurance, high mechanical flexibility along with high conductivity, and transparency. Up to this point, ITO, silver nanowires, Al-doped ZnO , carbon nanotubes, and organic materials have been utilized among which ITO still preserves its prevalence as a conductive electrode.⁸⁰ Hu and colleagues fabricated efficient flexible QDSCs by using conventional polyethylene terephthalate (PET)/ITO substrates.³ In this sense, the significance of further research on high performance and flexible SCs via using higher mechanical endurance of QD films has been emphasized. Up to now, certified efficiencies of 25.5 and 19.5% have been recorded on rigid and flexible conductive substrates, respectively.⁸¹

3.1.1.2. Electron Transport Layer (ETL). The presence of an ETL in a cell device is a prerequisite to obtaining a PSC with higher efficiency. An ETL expedites the assemblage of electrons and their journey to the corresponding electrodes.⁸²

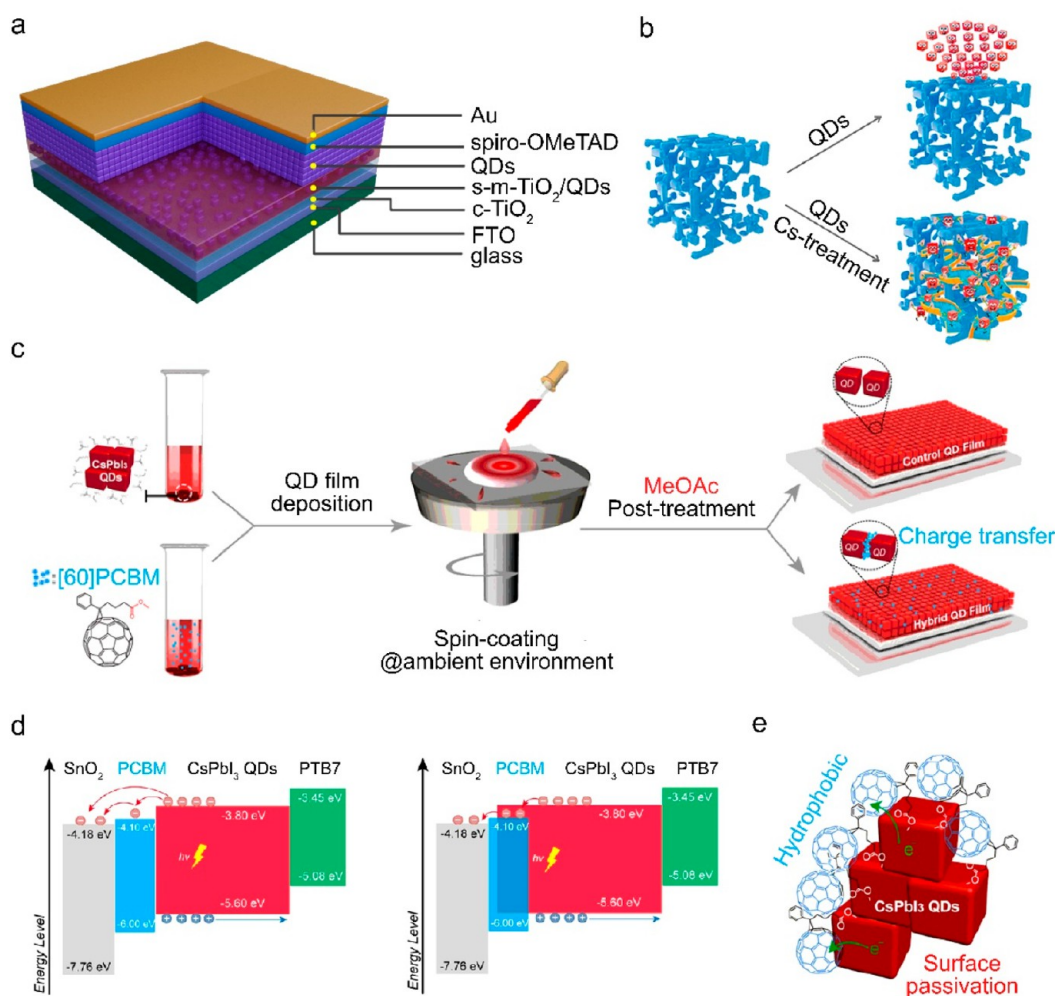


Figure 4. (a) Schematic view of the CsPbI₃ QD-based PSC with mesoporous structure. (b) Schematic of QD/TiO₂ mesoporous structure before and after Cs-treatment. Reproduced with permission.⁹⁷ Copyright 2020, The American Chemical Society. (c) Schematic of the fabrication of pristine, control, and hybrid CsPbI₃ QD films. (d) Band energy level diagram of control and target devices. (e) Schematic diagram of the hydrophobic PCBM passivating surface of CsPbI₃ QDs. Reproduced with permission.³ Copyright 2021, Nature.

Competent ETLs should hold a few features such as (i) decent electron mobility to ease electron transit within the layer, (ii) a broad bandgap to allow more light to travel through and thereby increase the light absorption amount by perovskites, and (iii) suitable energy levels with consistency between the conduction band minimum (CBM) and the valence band maximum (VBM) in connection with perovskite materials as light collectors, alleviating electron transit and obstructing hole motion through the perovskite layer. A compact ETL encourages the excited electrons from the active layer to the external circuit and at the same acts as a hole blocking layer (HBL) to obstruct exciting holes to FTO glass.⁸³ Likewise, compact ETL restricts backward electron transit from FTO to the active layer.⁸⁴ As a result, a compact ETL needs to consist of a structure that is regular, stable, as well as ultrathin. An ETL with a mesoporous structure in a PSC device accommodates hollow spaces within its framework, which is a path for perovskites to permeate. A mesoporous structured ETL expands the interfacial surface area between the active layer and ETM, providing facile charge transfer kinetics.⁸⁵ To prevent direct contact between the active layer and the conducting substrate, a compact ETL or so-called HBL is placed under the mesoporous ETL. Lacking the HBL gives rise

to back recombination of electrons and holes, and in turn, reduces values of open-circuit voltage (V_{OC}) and fill factor (FF) of the device.⁸⁶ So far, many inorganic compounds have been nominated as promising ETLs for PSC devices, some of which are TiO₂,⁸⁷ SnO₂,⁸⁸ and ZnO.⁸⁹ Various manufacturing techniques have been identified for the deposition of ETLs on the conducting electrode. Based on the utilized conducting substrate, ETLs can be obtained via high- or low-temperature processes.

3.1.1.3. ETL/QDs Chemical Interface Engineering. Engineering the active layer (perovskite) and correlated interfaces (active layer-ETM and active layer-HTM) are critical parameters to develop PSC devices with high performances. Nevertheless, while studies on the fine-tuning of the interfaces and their chemistry are quite scarce, a great deal of research has been conducted on the surface passivation of CsPbI₃ QDs utilizing Pb²⁺ or Cs⁺.^{90–92} To date, the most successful trial in terms of application of ETM in PSCs has been attributed to mesoscopic TiO₂ (m-TiO₂) thanks to its perfectly large interfacial contact with the active layer.^{93–95} Considering the fact that film growth takes place in a brief period, the homogeneous introduction of QDs into the mesoporous structure is not a straightforward task to fulfill. Therefore,

researchers have mostly used compact TiO_2 (c- TiO_2) in preference to m- TiO_2 to develop all inorganic lead halide QD-based SCs.^{90,96} In addition to this, m- TiO_2 films contain much larger irregularities on the surface as opposed to c- TiO_2 films that bring forth the formation of a QD capping layer, leaving negative effects on the interfacial properties of TiO_2 /QD. In this respect, proper modulation of both the ETM layer and ETM/QDs interface are the fundamental ingredients to attain α - CsPbI_3 QD-based PSCs with high performances.⁹⁷

Chen et al.⁹⁷ have studied the chemical interface of ETL/QDs by using the mesoporous TiO_2 /QDs layer. The results of current–potential (J – V) characteristics for both c- TiO_2 (~40 nm) and m- TiO_2 (~200 nm) based CsPbI_3 PQDSCs were examined. c- TiO_2 -based PSCs (c-PSCs) delivered a short-circuit current density (J_{SC}) value of $14.44 \text{ mA}\cdot\text{cm}^{-2}$ and a PCE of 8.77%, which are appreciably higher compared to those of m- TiO_2 -based PSCs (m-PSCs) (10.07 mA cm^{-2} , 5.01%), suggesting better charge split-up and improved carrier transit for planar structure. The lower performance of m-PSCs can be ascribed to the inhomogeneous distribution of QDs in the framework of m- TiO_2 ETM along with improper engineering of the m- TiO_2 / CsPbI_3 QDs interface.

Figure 4a,b demonstrates the schematic view of the CsPbI_3 QD-based PSC with mesoporous structure and the introduction process of QDs into the structure of the mesoporous layer that has been treated by Cs. Lack of Cs-treatment results in easy adsorption of CsPbI_3 QDs on the surface of the mesoporous layer, blocking m- TiO_2 porous structure and in turn discouraging migration of QDs into the mesoporous layer. On the contrary, treatment with a cesium acetate (CsOAc)/methyl acetate (MeOAc) solution lowers the adsorption energy of the QDs on the surface of TiO_2 , leading to the effective introduction of QDs into the mesoporous layer. In brief, the Cs-m- TiO_2 /QDs film exhibited a 2- and 3-fold higher rates of electron delivery compared to respective films of c- TiO_2 /QDs and m- TiO_2 /QDs. It is noteworthy that the observed irregularities on the surface of m- TiO_2 film were correlated with the spin-coating process used for its fabrication. Chen and colleagues⁹⁷ conducted research to flatten the m- TiO_2 film through a postdeposition process called the smoothing (s-m- TiO_2) approach by employing ethanol. All the PQDSCs developed by Cs-s-m- TiO_2 /QD films delivered PCE values beyond 13%, confirming the efficacy of the smoothing approach by ethanol.

Hu et al.³ succeeded to build a thin hybrid interfacial architecture (HIA) via incorporating phenyl-C61-butyric acid methyl ester (PCBM) into the structure of CsPbI_3 QD taking most of QDs' massive surface areas. Figure 4c depicts the related QD film deposition. The PCBM forms bonds with Pb^{2+} ions on the surface of QD by means of carboxyl groups leading to the creation of an exciton cascade between SnO_2 as ETL and the CsPbI_3 QD layer leading to enhanced generation of excitons at the QD/ETL interfaces. From Figure 4d, it can be seen that PCBM with sufficient electron affinity acts as an adjustment layer between the SnO_2 and the CsPbI_3 QDs, restricting charge carrier recombination between the two layers. Consequently, PCBM is interpreted as a road for swift electron delivery in QD devices. It is pronounced from Figure 4e that integration of PCBM with CsPbI_3 QDs at the ideal ratio leads to efficacious electron transfer at the interfaces. Recently, Lim and colleagues⁹⁸ have reported that TiO_2 nanoparticles can expedite the degradation of cubic-phase CsPbI_3 -PQDs due to their intense photocatalytic activity.

Hence, they developed chloride-passivated SnO_2 (Cl@SnO_2) QDs as the ETL with poor photocatalytic activity. Compared to nonpassivated SnO_2 , the passivated Cl@SnO_2 ETL exhibited more hydrophobic characteristics and an enhanced Fermi level, which contributes to the improvement of device stability and V_{OC} , respectively. Given these advantages, the PQDSC based on Cl@SnO_2 ETLs demonstrated an enhanced PCE up to 14.5% compared to that of TiO_2 -based control devices (PCE of 13.8%) and tremendously improved stability, maintaining 80% of the initial PEC, without encapsulation under 50% RH conditions for 8 h compared to the TiO_2 -based control device (maintaining 28% of the initial PEC, at the same condition).

An overview of recent works on the interface between ETL and the perovskite layer may serve as a prototype for future research. Zhang and colleagues⁹⁹ have also used carbon dots (CDs) to modify interfaces between metal oxide ETLs and perovskite active layers. On the report of this, CD modifiers can reduce the ITO work function allowing to significantly improve the performance of PSC devices. Zhong and co-workers¹⁰⁰ lately documented that tetrabutylammonium acetate (TBAAC) as a buffer layer between the SnO_2 ETL and CsPbI_3 interface not only improved the conductivity of ETL but also formed a 1D TBApI_3 layer between the ETL and the perovskite film resulting in enhanced stability and perovskite cell performance. Likewise, black phosphorus QDs have been employed as an additive component to a precursor solution of SnO_2 nanoparticles and it could effectively improve the performance of ETL in the PSC device.¹⁰¹

3.1.1.4. Active Layer. A perovskite layer—active layer—is the core of a PSC device, containing a photoresponsive material and responsible for encouraging exciton generation. PQDs exceed their bulk counterparts in many aspects. The merits that they offer include but are not limited to tunable bandgap through a quantum-confinement effect,²¹ multiple exciton generation,²² a PLQY of nearly one,^{21,23} and photon up- and down-conversion.²⁴ Thereupon, QDs are expected to surpass the Shockley–Queisser limit²⁶ and excel in the tandem PSCs. Furthermore, PQDs can be more fascinating owing to their solution processability at ambient temperature, providing a facile technique and dissociation of film deposition and grain growth.⁴⁶ PQDs with satisfactory endurance against defects are rising as potential materials to supersede conventional chalcogenide QDs in PV applications.²¹ Yet, both CsPbI_3 and FAPbI_3 QDs for instance are extremely vulnerable to exposure to wetness and polar solvents after synthesis treatment.^{6,46,102} The preceding setback creates complications in ligand modulation and thereby escalates the number of defects throughout the framework, restricting carrier transit and minimizing PCE values. Figure 5 shows the plot of the year versus the record PCE of QD-based SCs. Table 1 summarizes the chronological improvement of PQDSCs with respect to their device architectures and photovoltaic performance.

3.1.1.4.1. Surface Ligand Management of PQDs. It is widely believed that the first surface layer of a 10 nm PQDs contains approximately 20% of the atoms. Various factors affect the surface chemistry of semiconductor nanocrystals, including crystal facets, stoichiometry, organic ligands, and the ionic state. Therefore, to rationally improve and tune the photophysical properties of halide perovskite NCs, it is essential to understand their surface chemistry.¹¹⁹ As reported by earlier research, X-ray photoelectron spectroscopy (XPS)

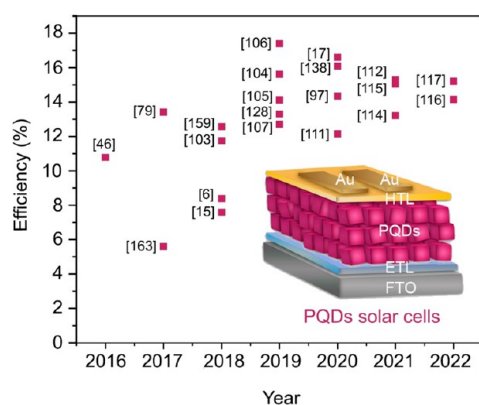


Figure 5. Plot of the record efficiency of PQDSCs each year.

analysis revealed that on the surface of the CsPbBr₃ NCs, the Br:Cs elemental ratio is ~ 5 and it decreases to ~ 3 in the bulk, indicating a surface excess of Br. However, for a CsBr termination, some of the surface Cs-sites could be occupied by alkylammonium cations, resulting in Br:Pb ratios exceeding 3 and Cs:Pb ratios of ~ 1 .¹²⁰ Moreover, the measured internuclear distances of H–Cs and H–Pb between dodecylammonium $-\text{NH}_3^+$ ligand protons and surface and subsurface Cs and Pb spins confirmed termination of CsBr with alkylammonium ligands which substituted into some surface A-sites.¹²¹

It is believed that the native ligands influence particle morphology, stability, and solution dispersibility, and they could passivate electronic trap states. It is also well-known that there are two kinds of native ligands on CsPbBr₃ NCs: long-chain primary alkylammonium ligands (oleylammonium) and long-chain carboxylate ligands (oleate). The maximum density of organic ligands has been demonstrated to be approximately

2–3 ligands per nm² for ligands with 0.3–0.5 nm² ligand footprints.¹²¹ It should be highlighted that even in nonpolar solvents, the native surface ligands are unstable, which results in poor stability NCs.¹²² It was determined by Brutchey et al.¹²³ that the native oleylammonium and oleate ligands can be exchanged with free alkylamines and carboxylic acids inside a solution. Inferred from this dynamic equilibrium of ligand, it is favorable for oleate to exchange with 10-undecenoic acid at room temperature or higher. On the contrary, with increasing temperatures, the exchange of oleylammonium with 10-undeceneamine becomes less favorable. A collection of amine ligands was examined for the exchange of native amine ligands of CsPbBr₃ NCs in order to effectively passivate the surface.¹²⁴ In this sense, the stoichiometric addition of strongly basic primary alkylamines (short C4–C8) could effectively bind to the surface and improve colloidal stability.

In 2020, Wang et al.¹⁷ introduced Cs_{1-x}FA_xPbI₃ without Br and volatile MA as a decent perovskite, demonstrating better moisture, thermal, and photostability. With reference to stability and charge transport traits, mixed-cation Cs_{1-x}FA_xPbI₃ QDs are more worthwhile than their pure ingredients—CsPbI₃ and FAPbI₃ QDs. The reason is that the resulting Cs_{1-x}FA_xPbI₃ QDs possess higher stability under ambient conditions or in contact with polar solvents in comparison with pure CsPbI₃ or FAPbI₃ QDs. Besides, they display extremely smaller trap densities and longer carrier lifetimes. The simultaneous introduction of FA and Cs generates entropic stability in the structure of perovskite under surrounding conditions.¹²⁵ In addition, the swift rotation of FA brings about intensified orbital overlap and simpler establishment of polaron, giving rise to lower nonradiative recombination and longer carrier lifetimes.¹²⁶ Furthermore, the enhanced light collection followed by greater J_{SC} is because of the wider spectral absorption as a result of the introduction of FA.

Table 1. Summary of the Device Architectures and Photovoltaic Performances of PQDSCs

Cell stack	PCE (%)	V_{OC} (V)	J_{SC} (mA/cm ²)	FF	Year	Ref.
FTO/TiO ₂ /CsPbI ₃ QD/Spiro-OMeTAD/MoO _x /Al	10.77	1.23	13.47	0.65	2016	46
FTO/TiO ₂ /AX coated CsPbI ₃ QD/Spiro-OMeTAD/MoO _x /Al	13.43	1.16	15.24	0.76	2017	79
FTO/TiO ₂ /CsPbI ₃ QD/Spiro-OMeTAD/MoO _x /Al	11.75	1.17	15.2	0.66	2018	103
ITO/SnO ₂ /FAPbI ₃ QD/Spiro-OMeTAD/Au	8.38	1.10	11.84	0.64	2018	6
FTO/TiO ₂ /α-CsPbI ₃ + FAPbI ₃ QD/PTAA/MoO ₃ /Ag	15.6	1.22	17.26	0.74	2019	104
FTO/TiO ₂ /CsAc passivated CsPbI ₃ QD/PTAA/MoO _x /Ag	14.10	1.248	14.96	0.75	2019	105
ITO/TiO ₂ /Cs _{0.25} FA _{0.75} PbI ₃ :CsPbI ₃ (1:3)/Spiro-OMeTAD/MoO _x /Al	17.39	1.20	18.91	0.76	2019	106
FTO/c-TiO ₂ /Cs-s-m-TiO ₂ /CsPbI ₃ QD/Spiro-OMeTAD/Au	14.32	1.06	17.77	0.75	2019	97
ITO/SnO ₂ /FAPbI ₃ CQD with ITIC/Spiro-OMeTAD/Ag	12.7	1.10	15.4	0.74	2019	107
FTO/TiO ₂ /CsPbI ₃ + PBDB-T QD/PTAA/MoO ₃ /Ag	13.8	1.22	15.1	0.75	2020	108
ITO/SnO ₂ /glycine-based PQDSCs/Spiro-OMeTAD/Ag	13.66	1.22	17.66	0.63	2020	109
ITO/SnO ₂ /Cs _{0.5} FA _{0.5} PbI ₃ QD-OR/Spiro-OMeTAD/Au	16.6	1.17	18.3	0.78	2020	17
ITO/ZnO/PTB7 Th:FOIC:PQDs/MoO ₃ /Ag	13.2	0.740	25.0	0.71	2020	7
FTO/NiO/FAI treated CsPbI ₃ QD/C60/ZnO/Ag	13.1	1.19	14.25	0.77	2020	110
FTO/c-TiO ₂ /m-TiO ₂ /CsPbI ₃ PQDs/Cu ₁₂ Sb ₄ S ₁₃ QDs/Au	10.02	1.04	18.28	0.52	2020	111
ITO/SnO ₂ /PCBM@CsPbI ₃ QD/PTB7/MoO ₃ /Ag	15.1	1.26	15.2	0.78	2021	3
FTO/Cl@SnO ₂ /CsPbI ₃ QD/P3HT/MoO _x /Ag	14.5	1.23	16.4	0.71	2021	98
FTO/TiO ₂ /CsPbI ₃ -Pe-CQD/Spiro-OMeTAD/MoO _x /Ag	15.3	1.27	16.1	0.74	2021	112
FTO/m-TiO ₂ /CsPbI _{2.4} Br _{0.6} QDs/Spiro-OMeTAD/Au	12.31	1.20	14.21	0.72	2021	113
FTO/TiO ₂ /GBA CsPbI ₃ QDs/Spiro-OMeTAD/Ag	13.2	1.25	14.9	0.71	2021	114
FTO/TiO ₂ /CsPbI ₃ QDs/Y6-F hybrid/thin QD/PTAA/MoO ₃ /Ag	15.05	1.26	15.81	0.75	2021	115
FTO/TiO ₂ /Pmim]I-CsPbI ₃ PQDs/Spiro-OMeTAD/MoO _x /Ag	14.14	1.20	15.65	0.75	2022	116
FTO/TiO ₂ /CsPbI ₃ QDs treated TPPI/PTAA/MoO ₃ /Ag	15.21	1.23	15.54	0.79	2022	117
FTO/TiO ₂ /P-N homojunction CsPbI ₃ QDs/PTAA/MoO ₃ /Ag	15.29	1.25	17.12	0.71	2022	118

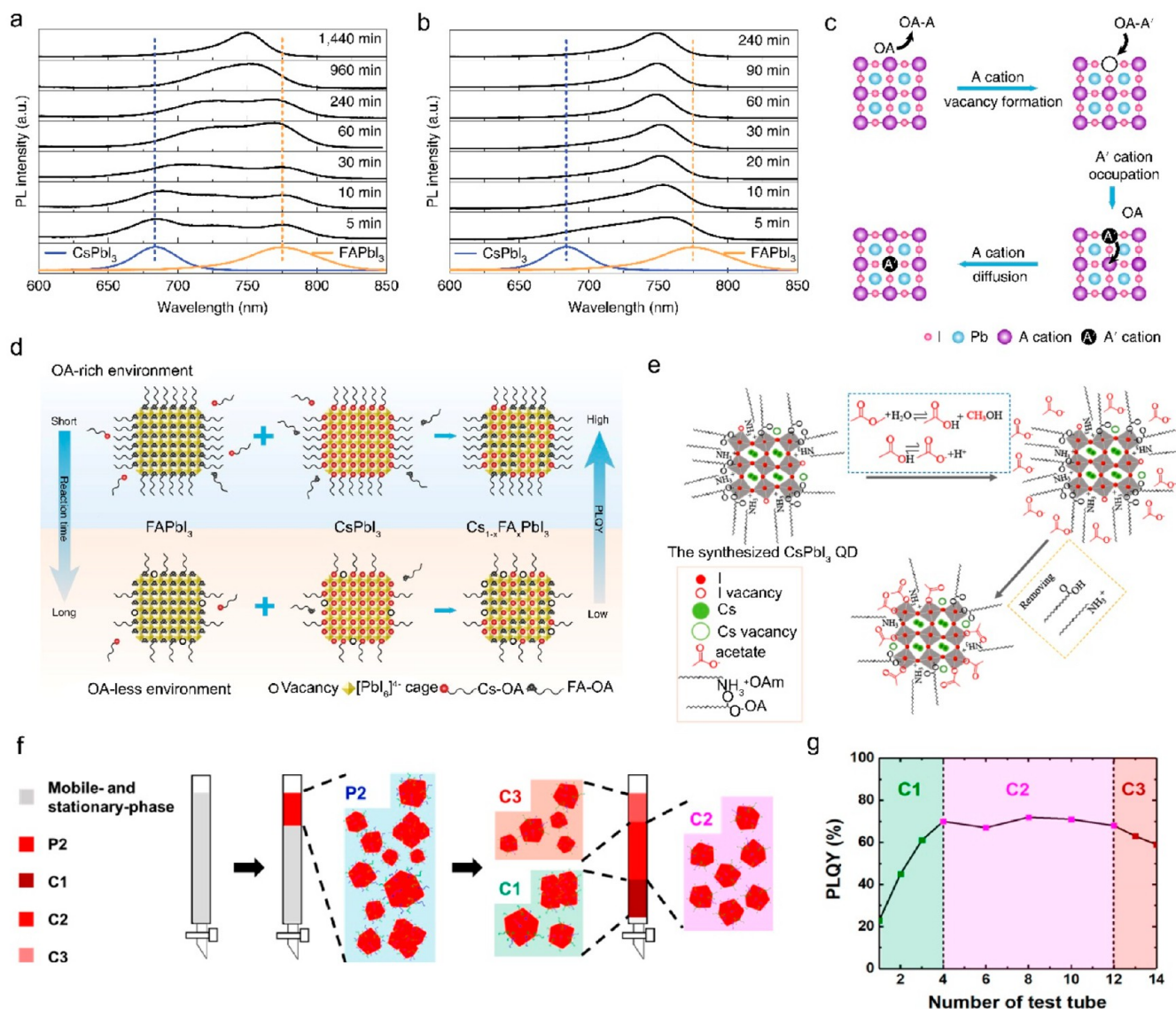


Figure 6. (a,b) Evolution of PL emission peaks with time to form Cs_{0.5}FA_{0.5}PbI₃ QDs by combining parent QD solutions that were purified twice (OA-less, a) and once (OA-rich, b), respectively. The bottom-most spectra in these two figures show the individual PL emissions from CsPbI₃ (blue) and FAPbI₃ (orange). The remaining emission spectra (black) are shown for the temporal evolution with time. Before mixing, they exhibit a PL peak at 683 and 774 nm for CsPbI₃ (blue dashed line) and FAPbI₃ (orange dashed line), respectively. (c) Proposed A-site cation-exchange reaction mechanism. Desorption of solvated cations from QDs leaves behind cation vacancies at the surface for the new cations to occupy. (d) Schematic illustration of cation exchange in different environments. Shuttling of FA⁺ and Cs⁺ cations between QDs is facilitated by the diffusion of Cs-oleate and FA-oleate in solution. As a result, the cation-exchange process under OA-rich conditions is promoted compared with that in the OA-less environment. Reproduced with permission.¹⁷ Copyright 2020, Nature. (e) Schematic illustrations of MeOAc Treatment on the Surface of CsPbI₃ PQDs. Reproduced with permission.¹²⁷ Copyright 2021, The American Chemical Society. (f) Schematic illustration of the size selection via GPC and (g) variation in solution-phase absolute PLQYs according to the elution sequence of the Pe-CQD eluate after GPC. Reproduced with permission.¹¹² Copyright 2021, The American Chemical Society.

The development and diffusion of A-site cation vacancies, which are of significant importance for inducing the cross-exchange of cations, are subject to the surface ligands. Wang et al.¹⁷ suggested that the oleic acid (OA) ligands can be utilized to remove obstacles by solvation of A-site cations of QDs including Cs-oleate and FA-oleate in the colloidal solution. They stated that once the original colloidal solutions (CsPbI₃ and FAPbI₃) were treated twice with MeOAc prior to mixing, a great portion of OA ligands was eliminated, leading to the generation of an OA-deficient environment (Cs_{0.5}FA_{0.5}PbI₃-QD-OD). To integrate the two distinctive photoluminescence (PL) peaks of CsPbI₃ and FAPbI₃ QDs peaks into a single and

wide peak, tens of hours were spent. The presence of a high kinetic barricade for cations to leave the initial QDs and incorporate them into other QDs is signified through the sluggish reaction. To this end, to produce an OA-enriched environment (Cs_{0.5}FA_{0.5}PbI₃-QD-OE), Wang and colleagues¹⁷ preserved more OA ligands in the original colloidal solutions. As a result, the two above-written PL peaks were swiftly integrated into one on account of mixing and the produced emission spectrum did not show alteration within 30 min, manifesting high-speed cation exchange completion (see Figure 6a,b).

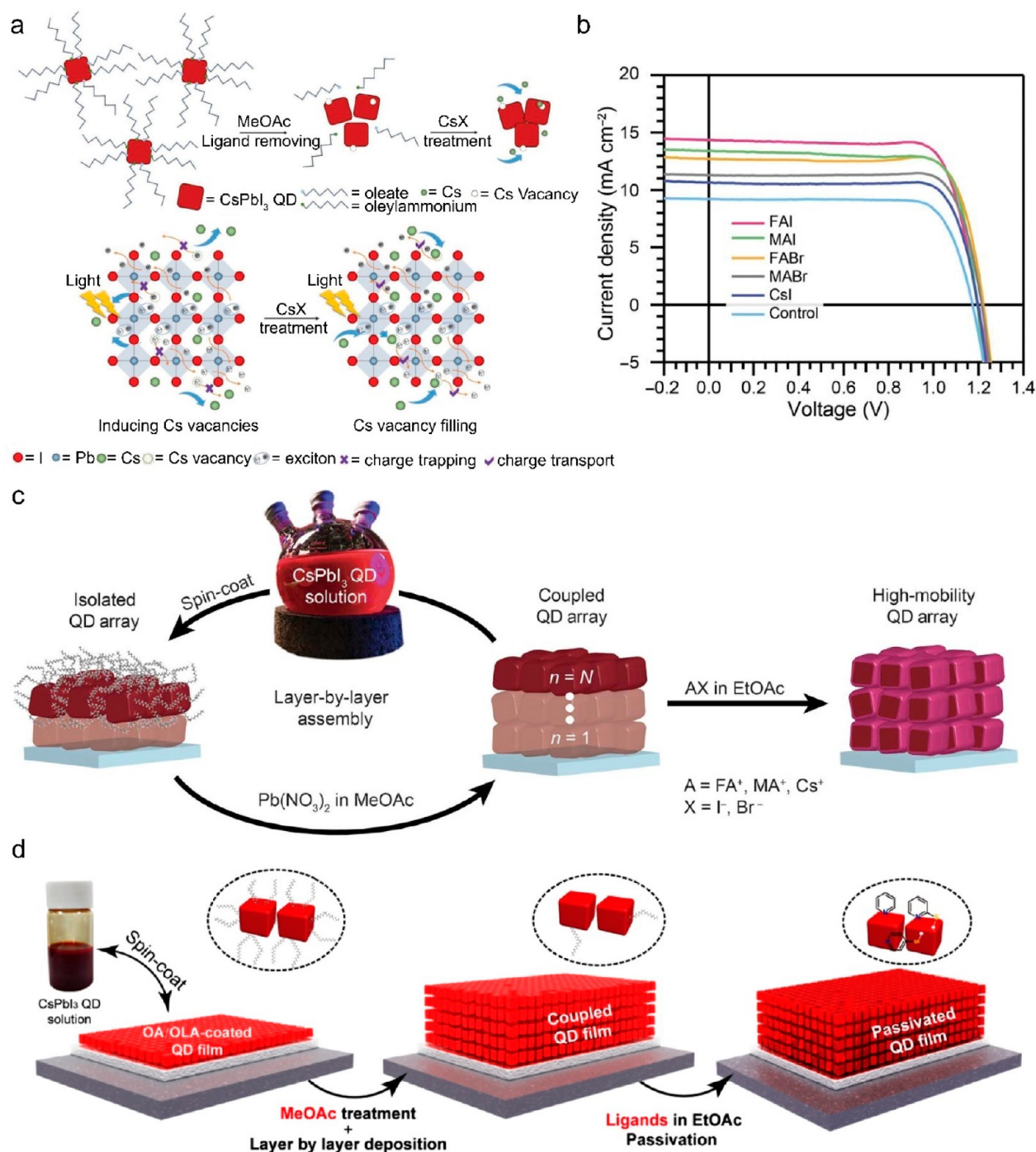


Figure 7. (a) Schematic illustrations of CsPbI₃ QD film deposition and CsX post-treatment process (upper part) and internal exciton generation, charge trapping, and transport process after ligand removing and Cs-salt post-treatment (bottom part). Reproduced with permission.¹⁰⁵ Copyright 2019, Wiley. (b) *J*-*V* characteristics of CsPbI₃ QD devices treated with FAI (pink), methylammonium iodide (MAI) (green), formamidinium bromide (FABr) (yellow), methylammonium bromide (MABr) (gray), cesium iodide (CsI) (dark blue), and neat EtOAc control (blue). (c) Schematic of the film deposition process and AX salt post-treatment. Reproduced with permission.⁷⁹ Copyright 2017, Science. (d) Schematic diagram of the fabrication of CsPbI₃ PQDSCs. Reproduced with permission.¹³³ Copyright 2020, The American Chemical Society.

The fundamental characteristic in ligand tuning is the significant difference that maintains high radiative performance by restraining the growth of surface defects in addition to the promotion of the cation-exchange reaction (shown in Figure 6c,d). In light of the results, the CsPbI₃-QD device revealed a *J*_{SC} of 15.4 mA cm⁻², a *V*_{OC} of 1.16 V, and an FF value of 53.9%, producing a PCE of 9.6%. Surprisingly, SCs based on Cs_{0.5}FA_{0.5}PbI₃-QD-OD yielded a low *V*_{OC} of 1.08 V and a *J*_{SC}

of 14.6 mA cm⁻², delivering a PCE value similar to the reference device—10.1%. On the contrary, the PCE values obtained from SCs based on Cs_{0.5}FA_{0.5}PbI₃-QD-OE were beyond 16% owing to significantly improved *J*_{SC}. To be specific, the values achieved from Cs_{0.5}FA_{0.5}PbI₃-QD-OE devices were as follows, a PCE of 16.6%, a *J*_{SC} of 18.3 mA cm⁻², a *V*_{OC} of 1.17 V, and an FF of 78.3%.

Since MeOAc is a well-known solvent in the ligand exchange of PQDs, the reproducibility of PQDSCs exploiting MeOAc treatment has been considered one of the bottlenecks for these cells. The importance of keeping long-chain organic ligands for preserving the stability of the crystal phase and getting rid of the ligands for high charge transport drive researchers to investigate the degree of MeOAc treatment on CsPbI₃ PQD solid films. In this regard, recently Han and colleagues¹²⁷ studied the effect of MeOAc treatment on PQDSC performance. Based on that, using several cycles of MeOAc treatment within a certain range they could eliminate the ligands and passivate the defects of the PQD film surface, thereby increasing the charge transport inside the film. But excessive MeOAc treatment can induce undesired phase transition and degradation of the device. In general, I- and A-site vacancies are the main factors in creating defects. Thus, Lewis base groups including carboxylate and thiol groups might be absorbed to the uncoordinated Pb atom formed due to the I vacancy, while Lewis acids such as the ammonium group could be adsorbed to the uncoordinated I formed due to the Cs vacancy (see Figure 6e). Keeping these in mind, it was proven that optimizing the MeOAc treatment could enhance carrier lifetime and reduce trap density throughout the QD film.¹²⁷ Kim et al.¹²⁸ improved the electronic coupling and photovoltaic performance of CsPbI₃ PQDs by using sodium acetate (NaOAc) in MeOAc solution to remove the long-chain oleate ligands. In the conventional method, acetic acid and methanol are produced as intermediate substances in the hydrolysis of MeOAc. Consequently, the metal hydroxide (i.e., Pb-OH) is formed from methanol produced during the process. In the light of experimental results, by efficient removal of long-chain oleate ligands using a solution of NaOAc in MeOAc (which results in the direct generation of OAc ions without forming protons and methanol), surface trap states originating from metal hydroxide formation could be minimized, leading to an improved device performance of 13.3%, compared to a lead nitrate-treated control device (with a PCE of 12.4%).

It should be overemphasized that polar-antisolvent-based purification induces agglomeration of PQDs after ligand management steps bringing about the irregular size distribution of QDs. Therefore, the polydispersity of PQDs has been also considered one of the bottlenecks for the PQDSC fabrication due to bandtail broadening originating from irregular-sized PQDs. It is well established that these irregular-sized PQDs could enhance the energetic disorder in the PQD films. Recently, Lim and co-workers applied monodisperse of PQDs in identical sizes and band gaps in the PQDSCs. They demonstrated that well-purified monodisperse CsPbI₃ QDs could be obtained via size selection by using gel permeation chromatography (GPC). These monodisperse QDs exhibited higher PLQY in comparison to irregular-sized PQDs (see Figure 6f,g). In this respect, the SCs fabricated by well-purified monodisperse CsPbI₃ QDs have encompassed the highest PCE of 15.3% amidst all inorganic PQDSCs reported so far.¹¹²

Xue et al.⁶ categorized the solvents into three grades based on the magnitude of polarity in order to lay down a practical process for surface ligand treatment. For this, “grade I” indicates those solvents with a strong polarity that are able to annihilate ionic bonds in FAPbI₃. On the other hand, “grade III” is denoted to those solvents containing the least polarity, unable to split the bond between the ligand and FAPbI₃.¹²⁹ In this regard, successful treatment of ligands without the annihilation of CQDs needs solvents with medium polarity

(“grade II”). Accordingly, water, methanol, ethanol, and isopropyl alcohol labeled as grade I are strong solvents; t-butanol, 2-pentanol, acetonitrile (ACN)/toluene, and EtOAc as grade II are solvents with medium polarity; chlorobenzene, toluene, octane, and hexane as grade III bear weak polarity. Xue and colleagues⁶ were able to design a method that gradually decrease the density of ligand in solution and the solid phase, while the unity of FAPbI₃ CQDs remained largely intact. The gradual reducing the polarity of a grade II solvent such as 2-pentanol in each run of surface ligand treatment played an essential role in this procedure.

3.1.1.4.2. Surface Passivation and Post Treatment of PQDs.
3.1.1.4.2.1. Cesium Cation (Cs⁺) Passivation. Previous reports on PbS QDs have proved that surface ligands can modify some extraordinary properties of QDs such as solution processability, electron coupling between QDs in film, trap densities, and long-run durability.^{24,68} Thus, the extensive realization of the ligand exchange process during manufacturing CsPbI₃ QDSC along with the acceleration of charge transit is still of substantial importance. Thus, the development of a practical approach for surface treatment of PQDs is a critical key. To achieve this objective, a straightforward strategy for surface passivation of CsPbI₃ QD films employing several Cs-salts (CsOAc, cesium iodide (CsI), cesium carbonate (Cs₂CO₃), and cesium nitrate (CsNO₃) was performed by Ling et al.¹⁰⁵ They substantiated that cesium salts can boost both atomic defects by filling the vacancies and electron coupling between CsPbI₃ QDs.

To conduct surface passivation, pure MeOAc is applied to play two roles: (i) to eradicate intrinsic ligands on the CsPbI₃ QD during its fabrication and (ii) to provide proper grounds for succeeding layer growth. Once the required CsPbI₃ QD film with a suitable thickness (300–400 nm) is achieved, it is then washed with a saturated Cs-salt solution in ethyl acetate (EtOAc). As it was confirmed by Ling et al.¹⁰⁵ different Cs-salts can enhance the device performance of CsPbX₃ PQDSCs. Amidst the above-written salts, one stands out, CsOAc. The QD film treated by CsOAc, displayed boosted carrier mobility besides reduced recombination thanks to both improved electron coupling and surface passivation. To this end, the PEC values secured by CsOAc-, Cs₂CO₃-, CsI-, and CsNO₃-treated CsPbI₃ QDSCs are as follows 14.10%, 13.14%, 13.74%, and 13.67%, respectively, outrunning the PCE value of the control device, 12.59%. From Figure 7a, the solution processable CsPbI₃ QDs contain long intrinsic ligands remaining from a spin coating that can act as insulators. MeOAc is applied to wash the QD layer and remove insulating ligands, turning it into a conductive layer. Cs atoms can be lost in the course of this process together with bound Cs-oleate, creating Cs vacancies in perovskite structure.^{130,131} For the case of the QD layer treatment with the CsOAc solution in EtOAc, further washing with the EtOAc reinforced electron coupling between QDs, which in turn led to higher carrier delivery in CsPbI₃ QD layers.¹³²

3.1.1.4.2.2. Post-Treatment. QDs enjoy the wide bandgap tunability and its associated energy levels. Moreover, about CsPbI₃, there is additional merit—that is, phase durability. Yet, QDSCs display insufficient V_{OC} values compared to their wide bandgap and unsatisfactory efficiency when it comes to carrier assemblage. These drawbacks can be solved to a great extent by coupling CsPbI₃ QD layers. The AX technique is a general strategy to adjust the electronic attributes of the CsPbI₃ QD films. In this respect, Sanehira and co-workers⁷⁹ exploited an

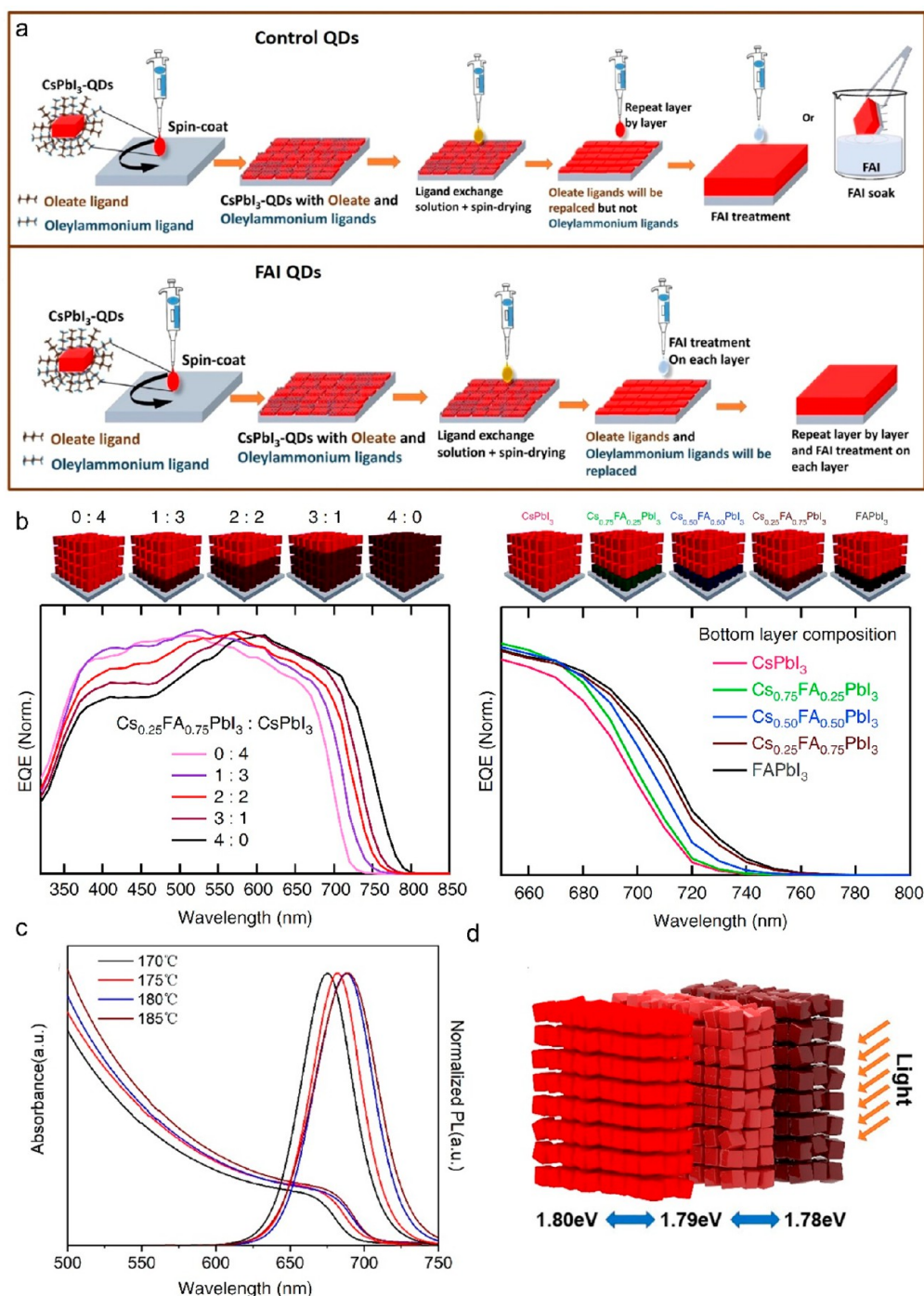


Figure 8. (a) Schematic diagram of solid-state ligand exchange of control QDs and FAI QDs. Reproduced with permission.¹¹⁰ Copyright 2020, The American Chemical Society. (b) EQE spectra of SCs with a varying thickness ratio of the Cs_{0.25}FA_{0.75}PbI₃ layer to CsPbI₃ layer in the perovskite QD absorber along with EQE spectra of SCs with different compositions of the bottom layer in the perovskite QD absorber where the thickness ratio of the mixed-cation Cs_xFA_{1-x}PbI₃ QD layers to CsPbI₃ QD layers is 1:3 in all cases. Reproduced with permission.¹⁰⁶ Copyright 2019, Nature. (c) UV-vis spectra and normalized PL spectra of the CsPbI₃ PQDs obtained at different synthesis temperatures along with (d) the GBA homojunction structure in SCs. Reproduced with permission.¹¹⁴ Copyright 2021, The American Chemical Society.

AX method (A-site cation halide salt) and confirmed the reliability of this method to modulate the electronic properties of QDs and subsequently improve carrier mobility.

Based on the results, in the case of addition of FAI to a CsPbI₃ QD film, although FA⁺ was clearly present in the treated films, the chemical composition of the QD core was unlikely to undergo drastic changes. It has been demonstrated that the AX salt-containing materials neither alloy with QDs

nor encourage grain growth. Instead, they coat the QD layer. Following deposition of the QD with an adequate thickness, it is afterward soaked in a saturated X salt solution in EtOAc. The comparison of *J*-*V* characterizations for devices with different AX salt treatments is illustrated in Figure 7b. Postsynthesis treatment that was carried out by each AX salt revealed a notable improvement in terms of the performance of the CsPbI₃ QDSCs compared with the control device,

manifesting the general applicability and efficacy of this technique. Figure 7c depicts a flow diagram, illustrating this process for film assembly. Amidst these AX salts, the device treated by FAI showed the highest PCE, 13.4%, and a J_{SC} of 14.37 mA/cm² which were considerably greater than those of the pure EtOAc control with a PCE and J_{SC} of 8.5% and 9.22 mA/cm², individually.

Khan and colleagues¹³³ reported the utilization of a *p*-mercaptopyridine (4-MP) ligand for post-treatment of PQDs and acquired not only boosted electronic coupling but also cubic phase robustness in post-treated PQDs in comparison with the ones treated with *o*-mercaptopyridine (2-MP) and pyridine ligands. They demonstrated that the residual oleic acid (OA) and oleylamine (OLA) native ligands can proficiently exchange with 4-MP ligands (schematic illustration shown in Figure 7d), which leads to the best efficiency of 14.25% due to the enhanced electronic coupling and fewer trap states. What is more, the intense anchoring strength of the bifunctional 4-MP linker leads to the improvement of the SC stability. The investigation manifested that the ligand surface coverage and their strength can be promoted by regulating the anchoring position, following an enhancement in the stability and performance of the PQDSC devices.¹³³ Analogous to this, Xue et al.¹⁰⁷ used conjugated small molecules ITIC in fabricating PQDSC to provide an additional driving force to separate charges more effectively. In this sense, boosted device performance approaching 13% in FAPbI₃ PQDSCs was recorded by removing the original insulating surface ligands of QDs and replacing ITIC in QD film by taking advantage of the layer-by-layer (LBL) deposition technique. Besides, an efficient solid-state-ligand exchange process was performed on a CsPbI₃ QD film via exploiting organic ligand triphenyl phosphite (TPPI) by Wang and co-workers.¹¹⁷ They proved that TPPI could balance the carrier delivery and thereby passivate the QD surface. In this sense, the CsPbI₃ PQDSC with a champion efficiency of 15.21% was achieved with an improved V_{OC} and a high FF.

3.1.1.4.2.3. Dual Passivation of PQDs. Jia et al.¹⁰⁹ studied the application of amino acids as dual-functional ligands that are able not only to enhance the defect passivation of PQDs but also to retain the cubic phase of PQDs. For the construction of the PQD film, first, a short ligand solution is prepared in the MeOAc solvent. Thereafter, for ligand exchange, the PQD solid film with OA and OLA ligands on the PQD surface is completely rinsed with a short ligand solution. Applying amino acids as dual-passivation ligands can passivate both the A-site (cesium) and iodine vacancies on the PQD surface, resulting in enhanced defect passivation. The results proved improvement of the charge transport property between individual PQDs through a short alkane chain of amino acids. In this respect, Jia and colleagues¹⁰⁹ constructed PQDSC devices containing planar structure to explore the PV characteristics of amino-acid-based PQD solid films. For the sake of comparison, conventional PQDSC was also developed by exploiting the lead(II) nitrate (Pb(NO₃)₂)-based PQD solid film as a light absorber in the device. From the results, the Pb(NO₃)₂-based PQDSC yields a J_{SC} of 16.30 mA cm⁻², a V_{OC} of 1.19 V, and an FF value of 0.60 with a high PCE of 11.69%, confirming the fact that the obtained results are analogous to or even better than the reported values for PQDSCs in the literature.⁷⁹ On the contrary, the PQDSCs based on glycine displayed 17.66 mA cm⁻², 1.22 V, and 0.63 for J_{SC} , V_{OC} , and FF, respectively, along with an improved PCE of about 13.66%.

3.1.1.4.2.4. Layer-by-Layer FAI Treatment. Shivarudraiah et al.¹¹⁰ studied LBL FAI treatment on inverted systems containing all-inorganic charge transport layers with the following arrangement: FTO/NiO_x/CsPbI₃QDs/C₆₀/ZnO/Ag. Compared to a typical single-step postdeposition FAI treatment, in this method, each layer is treated separately with FAI, which brings about notable PV performance. A thorough investigation conducted on the device configuration shows that the LBL FAI treatment approach discards intrinsic ligands including oleate and oleylammonium and offers better surface passivation and electron coupling. In this technique, a layer of QD solution is spin-cast onto the substrate several times. The chemical substitution of oleate ligands can be carried out via two sequential treatments including saturated salts of Pb(NO₃)₂ in MeOAc solution and AX salts. The Pb(NO₃)₂ in MeOAc solution is applied to treat control QD films in each layer individually, and subsequently, a single post-treatment is conducted through FAI in an EtOAc solution. Figure 8a depicts the relevant ligand exchange protocol. Shivarudraiah and colleagues¹¹⁰ proved the complete elimination of nearly all oleylammonium, oleate, or octadecene, confirming the substantial degradation of all intrinsic ligands for the FAI-QD films. As a result, it can be stated that the LBL FAI treatments appear to be quite useful not only to discard a great deal of carbonaceous materials but also to create a thick layer and produce better films. It should be noted that the FAI treatment does not alleviate the formation procedure of QDs alloy (Cs_xFA_{1-x}PbI₃) which is substantiated by the lack of noticeable change in the absorption and PL peaks. Consequently, the following statement can be drawn to attention that the effect of the LBL FAI process is restricted to improved surface passivation and QD coupling. Based on the results, control QDSCs delivered a V_{OC} of 1.17 V, a J_{SC} of 13.12 mA/cm², and an FF value of 0.752 with a PCE of 11.5%. Encouragingly, the FAI QD-based SCs displayed a V_{OC} of 1.19 V, a J_{SC} of 14.25 mA/cm², and an FF value of 0.776 with an enhanced PCE of 13.10%.

3.1.1.4.3. PQDs Heterostructure. Heterostructure-containing optoelectronics can remarkably regulate the energy positions of electrons and holes within the device.^{134,135} Not to mention that in PV devices, fine-tuning the energy band promotes the collection of photoinduced charge carriers.^{136,137} The role of solvents in PQDSCs is not to dissolve starting materials, but precisely, to form a suspension of QDs through their surface ligands under a colloidal solution. In a typical LBL technique, initially, the deposition of a layer of QDs is executed, followed by ligand removal, making the resulting layer nonsoluble in the nonpolar solvent, preparing fine conditions for succeeding film deposition without impairing the layer underneath.^{46,79,102} The active layer can be made up of either one type of QDs or a composition of various QDs, enabling the modification of the perovskite layer according to the required target across the film. For instance, Zhao et al.¹⁰⁶ conducted a spin-coating technique containing four courses, with which a 300 nm-thick layer was produced with respect to the concentration of QD and solution properties.

It was discovered that underlying film engenders a certain shift in the absorption onset of the composite. Besides, the effect of the thickness ratio of the Cs_{0.25}FA_{0.75}PbI₃ layer—composed of different compositions—to the CsPbI₃ layer was examined via the *J*–*V* characterizations. The results revealed that the device with a Cs_{0.25}FA_{0.75}PbI₃:CsPbI₃ of ratio 1:3 was named the best-performing device with a PCE of 15.52%. This

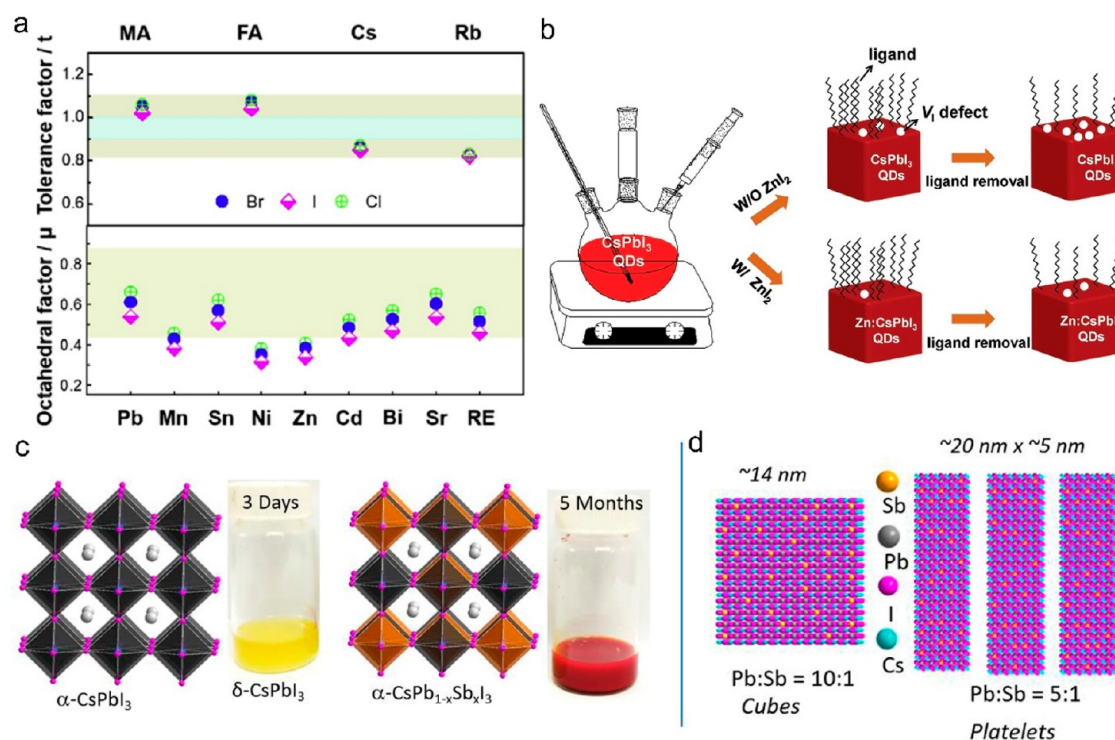


Figure 9. (a) Tolerance factor of 3D lead halide perovskite with general A-site cations, X-site anions, and octahedral factor of $[BX_6]^{4-}$ octahedra with possible B-site cations. Reproduced with permission.¹¹ Copyright 2019, Elsevier. (b) Schematic illustration of VI defect state control by ZnI_2 during synthesis and ligand removal processes of $CsPbI_3$ QDs. Reproduced with permission.¹⁴⁹ Copyright 2020, Wiley. (c) Atomic model of the crystal structure of α - $CsPbI_3$ and digital image of the nanocrystals of stored α -(cubic) $CsPbI_3$, which turned to yellow-colored δ -(orthorhombic) $CsPbI_3$ within 3 days along with atomic model showing that Sb(III) occupied the Pb(II) position in the α - $CsPbI_3$ crystal structure and Digital image of as-synthesized α - $CsPbI_3$ nanocrystals having $\sim 10\%$ Sb(III) incorporated and stored for 5 months (These nanocrystals were stored at a relative humidity of $44 \pm 6\%$). (d) Atomic models of cubes and platelets having 10 and 20% Sb (w.r.t. Pb) obtained from reaction with different Sb intakes. These percentages were calculated from EDS in the nanocrystals, but originally, the reaction intakes were 25 and 50% Sb with respect to Pb for cube and platelet formations. Reproduced with permission.¹⁵² Copyright 2019, The American Chemical Society.

outstanding performance stems from intensified J_{SC} followed by higher photoinduced carrier collection with this construction. As shown in Figure 8b, the external quantum efficiency (EQE) of cells is subject to the internal heterostructure interface position of the layers in which the collection efficiency was diminished for those devices including higher amounts of FA. Meanwhile, a systematic shift in the absorption onset is also caused by the composition of the bottom layer. What is more, it was found that device performance is quite similar for all following compositions $Cs_{0.75}FA_{0.25}PbI_3$, $Cs_{0.5}FA_{0.5}PbI_3$, $Cs_{0.25}FA_{0.75}PbI_3$, or $FAPbI_3$, when they are used as underneath layer in the absorber. The reason may lie in identical energy levels and charge split up in every one of the mentioned compositions.¹⁰⁶

The accomplishment of production of competent and durable single-junction PQDSCs is subject to meeting two requirements including (i) long-distance electron delivery and (ii) prolonged photon absorption. Li and co-workers¹⁰⁴ made an effort on the α - $CsPbI_3$ /FAPbI₃ bilayer structure to adjust simultaneously both light collection and the electronic traits. Based on the results, the best-performing title was devoted to the single α - $CsPbI_3$ QD device with a PCE of 12.3% and a J_{SC} of 13.80 mA/cm² comprising of four-layer α - $CsPbI_3$ QDs. At the same time, the single FAPbI₃ QD device comprising three-layer FAPbI₃ QDs displayed a PCE and J_{SC} of 12.0% and 15.97 mA/cm², separately. The higher J_{SC} of FAPbI₃-based devices, as opposed to α - $CsPbI_3$ -based SCs, is associated with the former's wider absorption. Conversely, the devices composed

of both α - $CsPbI_3$ and FAPbI₃ QDs demonstrated meaningfully enhanced performance. To this end, a device consisting of three layers of α - $CsPbI_3$ and two layers of FAPbI₃ QDs delivered a PCE, V_{OC} , and J_{SC} of 15.6%, 1.22 V, and 17.26 mA/cm², respectively, along with a relatively higher FF value of 0.74. Analogous to this work, Park et al.¹³⁸ reported a multinary PQD layer by combining $CsPbI_3$ and FAPbI₃ PQD layers to develop a heterostructured PQDSC. In this sense, a PCE of 16.07% with negligible hysteresis and significantly enhanced stability was obtained.

Yuan and colleagues¹¹⁴ benefitted from the gradient-band alignment (GBA) homojunction of PQDs in SCs. Consequently, $CsPbI_3$ PQDs with three different E_g values (by controlling the size of the PQDs) were used to fabricate SCs with a gradient-band homojunction structure (see Figure 8c,d). In this structure, due to the additional driving force for the electrons, the GBA could provide promoted charge extraction and carrier diffusion length in the PQD film. As a result, by reducing the potential loss of the SC, high efficiency of 13.2% could be achieved along with improved stability. In this heterojunction, the migration of doped ions could be also suppressed due to homojunction formed from same-composition PQDs.

Lately, Zhang et al.¹¹⁸ reported an internal P/N homojunction by using $CsPbI_3$ PQDs over 1 μ m-thick. In this context, to produce different carrier-type QD arrays, an organic dopant named F6TCNNQ was used in the arrays to transform the carrier-type from n-type to p-type. These different carrier-

type QDs were used in assembling P/N homojunction PQDSCs generating a boosted efficiency of 15.29%.

Hybrid PQD/nonfullerene molecule SCs have recently attracted great attention; however, their construction is still challenging. Yuan and co-workers¹¹⁵ reported hybrid SCs with superior PV performance consisting of CsPbI₃ PQDs and Y6 series nonfullerene molecules. In this work, CsPbI₃ QDs and nonfullerene molecules were used as a hybrid active layer. It was proven that the nonfullerene molecules not only could increase the charge transfer and extraction in the hybrid active layer by introducing a type-II energy alignment but also could passivate the QDs by diminishing surface defects and energetic disorder. Based upon the results, the high-performance hybrid SC with an efficiency of 15.05% was acquired, proving the rational engineering of the solution-processable hybrid active layer to adjust efficient optoelectronic devices.

3.1.1.4.4. PQDs Compositional Engineering. Although the phase stability of perovskites has been improved in the form of QDs compared to their bulk counterparts,¹³⁹ under operation, the stability of PQDSCs has been considered one of the bottlenecks for the PV devices. To circumvent this issue, we witnessed a lot of efforts by using fundamental strategies including compositional engineering, QD surface treatment, and optimization of the SC layered structure. Recently, compositional engineering has aroused great research interest in Cs-based PQDs due to its enormous capacity to tune the structures to avoid the formation of undesirable phase transitions. It is widely postulated that the stability of the SCs can be enhanced by modifying the perovskite composition to receive satisfactory tolerance factors. For the ABX₃ perovskite structure, appropriate doping of the A-site, B-site, and X-site can stabilize the crystal structure and improve the optoelectronic properties of the device. In the case of A-site doping, filling the Cs vacancies on the surface of QDs could promote the stability of perovskite structure in the air environment.¹⁰⁵ On the contrary, B-site doping could strengthen the structure along with adjusting the electronic properties of QDs. Besides, as the B-site cation is Pb²⁺, partially replacing Pb with a proper atom could also diminish nonradiative recombination, improving the device performance.¹⁰⁵ To obtain desired tolerance factor, the ionic radii of A-site, B-site, and X-site ions must be correlated to the Goldschmidt tolerance factor in the following relation:

$$t = r_A + r_X / \sqrt{2(r_B + r_X)} \quad (1)$$

where r_A , r_X , and r_B stand for ionic radii of A-site, X-site, and B-site ions, respectively. It has been demonstrated that in the view of SC with superior performance, the ABX₃ should be in a cubic phase structure consisting of corner-sharing [BX₆] octahedra with the A cation staying in the 12-fold coordination sites surrounded by the cube of eight octahedra.¹⁴⁰ For the stable cubic perovskite, the Goldschmidt tolerance factor, t , should be between 0.9 and 1.¹⁴¹ It should be noted that CsPbI₃ QDs hold a t value as low as 0.8. For this, the rational engineering of ionic radii—that is, diminishing the r_B and enlarging the r_A is accepted as an effective scheme to obtain a reliable tolerance factor. Likewise, the octahedral factor, μ , is another critical parameter to evaluate octahedral stability that is defined as the following:¹⁴²

$$\mu = r_B / r_X \quad (2)$$

To achieve the stable octahedral [BX₆]^{4−}, the required μ should be between 0.442 and 0.895. Figure 9a depicts some of the reported t and μ values for A-site and B-site doping in the PQD structure. Therefore, for B-site doping, satisfying the t along with μ is essential to meet the requirements of getting a stable cubic structure.

It should be stressed that although FA⁺ and MA⁺ as A-site candidates have larger ionic radii than Cs⁺, producing a highly anisotropic strain of cubic phase lattice (therefore creating deviation from the equilibrium interplanar distances)¹¹ along with the volatile nature of these cations are the major drawbacks of replacing the A-site with FA⁺ and MA⁺ cations.¹⁴³ The unsatisfactory efficiency and poor stability of organic–inorganic PQDSCs owing to the preceding bottlenecks have made researchers move to and adopt using all-inorganic CsPbI₃ QDs. Thereby, B-site doping serves as a prototype for future research in the development of all-inorganic CsPbI₃ QDs.

In addition, in the case of B-site doping, replacing Pb with a tiny amount of B-site dopant with a smaller ionic radius than that of Pb²⁺ can alter the phase structure and enhance optoelectronic properties.¹¹ Zou et al.¹⁴⁴ investigated Mn doping in CsPbBr₃ structure by DFT and proved that the much shorter Mn–Br bonds compared with Pb–Br bonds cause a lattice distortion that leads to better thermal stabilities of target PQD. Analogous to the Zou et al. work, Akkerman et al.²¹ showed that shorter Mn–I bonds in Mn-doped CsPbI₃ are responsible for the lattice contraction along with the negligible variation of the electronic structures at the band edges. Consistent with the tunable bandgaps, B-site doping could shift the PL position. In this way, proper doping can also boost the PLQY. In fact, B-site doping could regulate the light-emitting aspects of PQDs on account of the critical status of Pb in the PQDs' structure. It is also noteworthy to underline that according to eq 1, to increase the t value, replacing X-site with smaller elements e.g., Cl or Br would be even more beneficial. Nevertheless, these alloying elements induce an unintended blue shift in the absorption edge due to the bandgap broadening.¹⁴⁵ As a result, as a paradigm for future research of stable PQDSCs, developing new strategies toward improving the perovskite phase stability without increasing the bandgap via doping the B-sites with smaller metal cations is essential.

To date, several metal ions have been introduced to replace Pb²⁺ in PQDs including Mn²⁺,¹⁴⁶ Sn²⁺,¹⁴⁷ and Sr²⁺.¹⁴⁸ Zhang et al.¹⁴⁹ reported Zn-doped CsPbI₃ QDs exploiting ZnI₂ as the dopant with enhanced thermodynamic stability. Based on the results, the additional iodine ions could reduce the halogen vacancies on the QD surface (schematic illustration shown in Figure 9b), leading to diminished nonradiative recombination and improved PLQY. Furthermore, the Zn-doped CsPbI₃ PQDSC exhibited an enhanced efficiency from 13.98 to 16.07% compared to the undoped device. Analogous to this work, Bi and colleagues¹⁵⁰ reported CsPb_{1−x}Zn_xI₃ QDSCs with improved efficiency and stability. Behera et al.¹⁵¹ reported Ni(II) ion as a dopant in red-emitting CsPbI₃ QDs structure with a maximum PLQY of 81% and the stability recorded for 2 months. Liu et al.¹⁴⁷ also reported that CsSn_{1−x}Pb_xI₃ QDs alloy with superior stability to the CsPbI₃ and CsSnI₃ QDs parent structures.

Heterovalent B-site doping has been also used to enhance the optoelectronic properties of PQDs along with stabilizing the cubic phase at room temperature. Bera and colleagues¹⁵²

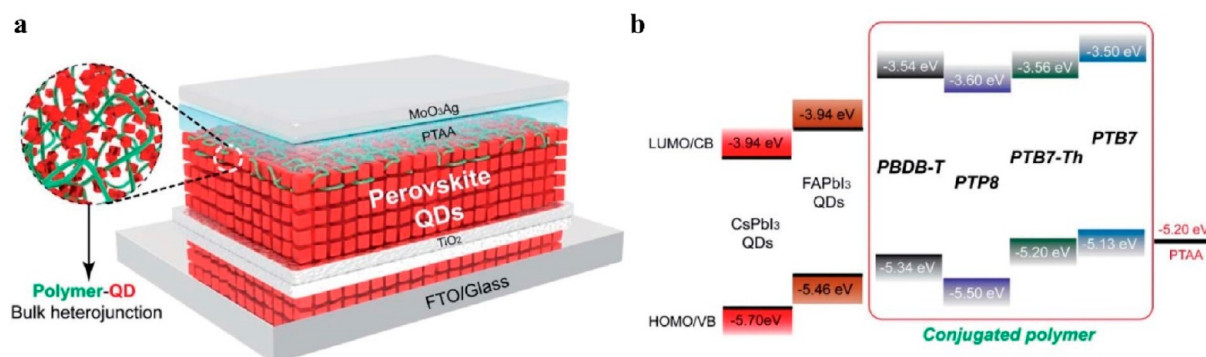


Figure 10. (a) Device structure of PQDSCs with polymer-QD BHJ connecting layers, and (b) energy levels of PQDs, conjugated polymers, and PTAA in this work. Reproduced with permission. Copyright 2020, The Royal Society of Chemistry.

doped CsPbI₃ QDs via a tiny amount of Sb³⁺ ions to obtain the phase and optical stability. The phase stability of Sb-doped α -CsPbI₃ QDs after 5 months without turning to yellow δ -CsPbI₃ is shown in Figure 9c. Likewise, for heterovalent Sb doping, the atomic model is shown in Figure 9d. Based on the model, cubes were formed when Sb(III) was assumed to be 25% of Pb; while adding a higher amount of Sb(III) resulted in platelets. To further study, the Yb³⁺ doping strategy has been also proposed by Shi and co-workers¹⁵³ to improve device efficiency and stability with reduced defects and trap states. Ghosh and colleagues¹⁵⁴ reported CsPbBr_{1.5}I_{1.5} QDs doped with monovalent Ag⁺ ions. The authors claimed that partial replacement of Pb²⁺ by Ag⁺ not only maintained the crystal structure but also reduced intrinsic defects, resulting in a long carrier lifetime. In this sense, the Ag-doped PQDSC with an increase in efficiency up to 20% and the device performance for at least 24 days with only a 5% drop in efficiency was achieved.

3.1.1.5. Hole Transport Layer. There are a few features that describe an excellent HTL, including but not limited to fast hole mobility, high conductivity, suitable alignment with the valence band of perovskite, as well as decent durability. HTLs in PSCs can be constructed from a wide spectrum of materials such as organics, inorganics, and polymers.¹⁵⁵

3.1.1.5.1. Organic and Polymer Hole Transport Materials. Small organic materials that are readily soluble in organic solvents are considered promising HTMs since they provide better contact with the active layer regardless of the morphology of the active layer. Knowing that organics are mostly composed of earth-abundant elements such as carbon, hydrogen, and nitrogen, they are potentially cheap materials. Spiro-OMeTAD is a small organic material that is commonly used as the HTL in PSCs. The application of Spiro-OMeTAD as an alternative to the liquid electrolyte in PSCs does not merely extend the device lifespan, but at the same time enhances its efficiency.^{61,156} Even though the triarylamine and spiro structure hamper the crystallization process in this layer, it suffers from both low mobility ($<10^{-4}$ cm² V⁻¹ s⁻¹) as well as conductivity (10^{-5} S cm⁻¹), requiring additional species to encourage the mentioned characteristics. Thus, far, different small organics have been identified as HTMs including pyrene derivatives, thiophene derivatives, triptycene derivatives, triazine derivatives, and triphenylamines derivatives, some of which can exhibit a performance equal to Spiro-OMeTAD.¹⁵⁷

The mobility and conductivity properties of polymeric HTMs surpass those of organics. Nonetheless, identifying a legitimate organic solvent that is able to dissolve the polymeric layer without affecting the perovskite layer is challenging. Due

to the larger size of polymeric materials, the contact between active and hole transport layers particularly in the case of PSCs—consisting of a mesoporous electrode—is insufficient.¹⁵⁸ It should be noted that polymeric HTMs are intrinsically more stable compared to small organics. Poly[bis-(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) and poly(3-hexylthiophene-2,5-diyl) (P3HT) have been the most commonly employed polymeric HTMs. Yet, their stability demands further improvement.¹⁵⁷ Yuan et al.¹⁵⁹ have employed a series of dopant-free polymeric HTMs in CsPbI₃ PQDSCs. Using QD/polymer interfaces, efficient charge extraction is achieved, and device instability (due to oxidation processes required by conventional Spiro-OMeTAD) is avoided. In this sense, the PQDSC with an efficiency of 13% and a very low energy loss of 0.45 eV was generated.

3.1.1.5.2. Inorganic Hole Transport Materials. Considering that the crystallization of the HTL lowers the contact at the interface of perovskite/HTL, the amorphous nature of inorganics seems such a necessity. Following depositing these materials on the substrate, some of them require a sintering process. It is worth noting that high-temperature heat treatment can destroy perovskite materials; on the flip side, the volatile organic compounds in the HTL pose a challenge regarding their stability issue. Thus, some of the inorganic HTMs are inevitably utilized in the inverted configuration PSCs. The higher durability of inorganic HTMs is their unique merit which is not the case for both organics and polymers. Two quintessential examples of inorganic HTMs can be named as NiO_x and CuSCN bearing high stability as well as performance. NiO_x which has been used in the inverted configuration PSCs demands a temperature as high as 600 °C during its heat-treatment process. Once it is assembled with the ZnO as the ETL in PSCs, the efficiency of the device is sustained for 60 days under light ambient conditions.⁷⁶ Liu et al.¹¹¹ developed PQDSC having the structure of FTO/c-TiO₂/m-TiO₂/CsPbI₃ QDs/Cu₁₂Sb₄S₁₃ QDs/Au, to use Cu₁₂Sb₄S₁₃ QDs as HTL to increase device stability. In light of the results, Cu₁₂Sb₄S₁₃ QD-based SCs generated a PCE of 10.02%, approaching the PCE of spiro-MeOTAD-based PQDSCs (12.14%). What is more, by improving the light absorption and the hole extraction ability of Cu₁₂Sb₄S₁₃ QDs, a J_{SC} of 18.28 mA/cm² was reached, and after 360 h of storage in ambient air, the PQDSCs retained 94% of their initial PCE.

3.1.1.6. QD/HTL Chemical Interface Engineering. Because of the huge difference in surface energy of QDs and polymers, the interfaces of PQDs and the organic HTL resulting from consecutive deposition processes are not perfect for efficacious

charge transport.¹⁶⁰ For this reason, fine-tuning the interfaces of QD/HTL is a feasible way to promote the charge dynamic process and thereby intensify the PV performance. Ji et al.¹⁰⁸ synthesized a solution-based conjugated polymer-PQD hybrid BHJ layer to achieve the preceding purpose (see Figure 10a). A proper selection of polymer brings about energy level hierarchy as QD/polymer/PTAA which in turn promotes charge carrier transport and minimizes recombination at PQD/HTL interfacial spots.¹⁶¹ The deposition of a conjugated polymer-QD hybrid layer on the QD film was carried out prior to spin-coating the organic HTL. The activity of CsPbI₃ and FAPbI₃ PQDSCs was explored utilizing different conjugated polymers (PBDB-T, PTP8, PTB7-Th, and PTB7).^{159,162} Energy level alignment on the interface of QD/polymer is an essential parameter, affecting carrier recombination loss and thereby improving carrier transport. Figure 10b displays a thorough comparison of the energy levels of PQDs and conjugated polymers as well as the HTL material PTAA. Ji et al.¹⁰⁸ selected conjugated polymers with distinctive HOMO energy levels positioned between the valence band of PQD and the HOMO of PTAA. CsPbI₃- and FAPbI₃-based PQDSCs deliver respective PCEs up to 13.1% and 11.6%. Interestingly, the addition of a layer of PBDB-T/QD BHJ promoted the PCEs of CsPbI₃ and FAPbI₃ PQD-based to 13.8% and 13.2%, separately. This betterment can be attributed to the improved J_{SC} value, stemming from the reduced carrier recombination at the interfaces. The charge-harvesting performance of the polymer-QD hybrid layer adjusted PQDSCs and demonstrated a noticeable improvement as against the control original PQD devices. It is postulated that a conjugated polymer such as PBDB-T is vital for the charge carrier mobility in the PQD in general and in particular at interfaces of PQD/HTL, suggesting effective charge separation and diminished charge recombination.

3.1.1.7. Back-Contact Electrode. Lastly, the finishing step involved in the fabrication of PQDSC is the deposition of metallic electrodes via thermal evaporation. Au^{6,111,163,164} and Ag^{16,107,109,110} electrodes are suitable candidates employed for the collection of holes. The metal oxides in combination with low work function metals such as MoO_x/aluminum^{46,79,103,106} and MoO₃/Ag^{3,5,104,105,108} are the other alternatives used for the back-contact electrode.

3.1.2. Stability Improvement. As of today, the highest efficiencies obtained for hybrid organic–inorganic halide perovskites are for those with the poorest stability, raising the most formidable challenge: device stability.¹⁶⁵ Although relative efficiencies to commercial silicon PV cells have been acquired for PQD cells, still comparing the lifespan of the former roughly 25 years with the latter from a few days to months simply indicates it is fairly difficult to fully supersede silicon PV cells with PQD cells. The components of PQDSCs seem to deteriorate immediately, which eventually leads the device to fall apart mainly because of the light-induced volatile organic components of the various layers. Several factors such as oxygen, moisture, temperature, illumination, biasing, and UV are affecting the stability of PQDSCs. Despite the improved device stability until now, it still has space to grow and be comparable to previously reported PbS-based QDSCs and some PSCs,^{166,167} implying that more investigations are required to excel the long-term performance of SCs for large-scale applications.

Up to this point, there have been some successful constructions of PQDs with high PLQYs such as MAPbX₃,

FAPbX₃, and CsPbX₃.^{103,168} Nonetheless, MA-based LHP QDs have low chemical tolerance because of their low formation energy, bringing forth their readily degradation in the course of separation and purification treatments.^{169,170} Considering the metastable form of the 3D phases of Cs- and FA-based LHPs under ambient conditions, downsizing the crystallite size from bulk to QD regime can indeed strengthen their tolerance.^{6,171} Cs-based LHPQDs, however, fail to be stable enough after a few weeks owing to the small-sized Cs⁺.¹⁷¹ In principle, phase transformation of CsPbI₃ QD films takes place in a matter of days to weeks, suggesting its close correlation with surface treating as well as storing conditions.^{79,172} Albeit the CsPbI₃ QD-based devices deliver insignificantly higher PCE value than that of FAPbI₃ QD-based SCs, the latter outweighs the former in terms of long-run stability.^{103,170,171}

Pan et al.¹⁷³ investigated the photo- and air-tolerance of an inorganic–organic hybrid ion pair as the capping ligand which showed the superior durability potential of CsPbBr₃ QDs with high PLQY. Dastidar and co-workers¹⁷⁴ proved that incorporation of chloride into CsPbI₃ by its maximum capacity of solubility can stabilize the α -phase. Moreover, Swarnkar et al.⁴⁶ witnessed the QD-triggered phase stabilization of α -CsPbI₃ with the help of two well-known ligands, OA and OLA. Wang et al.¹⁷⁵ used bis(2,2,4-trimethylpentyl)phosphinic acid (TMPPA), confirming its potential to replace OA to fabricate α -CsPbI₃. Considering numerous reports, the instability of α -CsPbI₃ still lingers. Although the chemical removal of surface ligands stabilizes the α -CsPbI₃ from the device's point of view, it ends up with phase degradation. In contrast, α -CsPbBr₃, an alternative absorber is a substantially stable phase. Pb-based perovskites are notoriously famous for their toxicity. The decomposition of these materials into Pb(OH)I complex and HI can be the reason behind their toxic nature. In addition, decomposition of the organo-halides can be accelerated once I[−] ions migrate in perovskite. Theoretical research corroborates that replacement of I[−] with Br[−] ions by 50% not only alleviates the diffusion barrier but also stabilizes the perovskite.¹⁷⁶ CsPbBr_{1.5}I_{1.5} constructed by Ghosh et al.⁵ was proven to be the best of all among halide compositions in terms of long-run PV performance with an infinitesimal drop in the PCE value. Accordingly, the CsPbBr_{1.5}I_{1.5} QDs, showed a longer lifespan for charge carriers and thereby higher assemblage efficiencies prior to recombination.

The presence of defects in the crystal lattice of perovskite gives rise to device instability.¹³² Ling et al.¹⁰⁵ employed CsAc to treat CsPbI₃ after synthesis. They stated that post-treatment results in stabilization of CsPbI₃ crystal lattice as well as mitigation of vulnerability of QDs against humidity as a consequence of filling of Cs vacancies. The device that underwent post-treatment with CsAc preserved 87% of its original PCE after its storage for a week in N₂, while the control device displayed 64% of its initial PCE under the same conditions. This finding suggests that Cs vacancies are an influential factor in chemical and/or structural degradation besides acting as charge trap and recombination sites. Wang and colleagues¹⁷ explored the long-run durability of the CsPbI₃ QD-based SCs under exposure to atmospheric conditions accompanying a relative humidity of 50–70% following the cation-exchange method. They concluded that Cs_{0.5}FA_{0.5}PbI₃ QD-based devices maintained 97% of their initial performance after storing for 20 days. The formation and migration of ionic defects inside the halide perovskite lattice are the major

contributors to ultimate device degradation.^{143,177} Therefore, this conclusion can be drawn to attention that superb device tolerance of $\text{Cs}_{1-x}\text{FA}_x\text{PbI}_3$ QD films under exposure to light is mainly due to appreciable suppression of defects in the alloy. It was stated by Jia et al.¹⁰⁹ that higher long-run performance is on account of the presence of denser hydrophobic glycine on the surface of PQD, offering better surface passivation and diminishing surface defects, as a result, the interaction between atmospheric vapor and PQDs is avoided.⁴⁶ Finally, Liu and colleagues¹¹³ also reported that Br doping can enhance the tolerance factor of the CsPbI_3 structure and consequently improve phase stability. From the results, the $\text{CsPbI}_{2.4}\text{Br}_{0.6}$ PQDSC with the PCE of 12.31%, is close to that of CsPbI_3 -based PQDSC (14.13%) and may retain 87% of its original efficiency after 15 days of storage in ambient air.

Even though colloidal PQDs exhibit significantly improved phase stability at room temperature in comparison to bulk counterparts,¹⁷⁸ the stability of PV devices based thereon under operation remains one of the most pressing research challenges facing the perovskites field. A significant amount of effort has been invested by the scientific community in order to address this issue. To this end, three commonly used fundamental strategies have been employed to improve the stability and durability of PQDSCs. In the first one, the focus is on the fabrication of PQD materials through compositional engineering and/or doping. The second strategy involves the surface treatment of QD films, and the third one focuses on optimizing SCs to protect the active absorber perovskite layer, including optimizing the transporting layer and deposition procedures.¹⁷⁹

As discussed in the PQDs compositional engineering section, Cs-based PQDs are prone to phase transitions due to their low structural tolerance factor. As a result, by adjusting the composition of perovskites to obtain appropriate tolerance factors, the stability of SCs can be improved. In this sense, a compositional engineering approach relaxes the lattice strain and suppresses phase transitions by controlling the A-site, B-site, and X-site atoms (see Section 3.1.1.4.4). While A-site doping could fill the Cs vacancies of QDs on the surface and improve the stability of perovskite structure in the air environment,¹⁰⁵ B-site doping could promote the stability of the structure along with adjusting the electronic properties of QDs.¹⁰⁵

Not to mention that, under operating conditions, light illumination and changes in the electric field cause ions to migrate toward the interfaces, resulting in a decrease in device stability. As a means of suppressing migration, it is necessary to reduce the defects and increase the vacancy formation energy. Since halide anions have a lower activation energy than A-site and B-site cations, they migrate more frequently. It is a well-known fact that substituting some portion of I anions with the smaller-size Br or Cl is another way to increase the tolerance factor (eq 1). Nevertheless, the Br and Cl alloying gives rise to an undesired blue shift of the absorption edge and an enlargement of the band gap.¹⁴⁵

The surface treatment of QD films is another important strategy to enhance the stability of PQDSCs. Although QDs are capped by ligands, they are largely removed during the purification and fabrication stages of the device, resulting in surface vacancies and QD agglomeration. These surface ligands, in addition to improving stability, also play an important role in the transport of photogenerated carriers. As compared to thin-film base devices, the relatively low J_{SC} of

CsPbI_3 PQD is mainly due to inefficient carrier transport. To achieve desired charge transport, following hot-injection synthesis, QDs capped by ligands with long alkyl chains should be partially removed. The antisolvent MeOAc is widely used in the purification of CsPbI_3 ligand PQDs and the reduction of the surface-bound oleate. Surface ligand density is critical to determining the balance between carrier transport and PQD phase stability. In specific, if the ligands are too dense, interdots charge hopping will be hampered, whereas too few ligands will prevent cubic CsPbI_3 PQDs phase stabilization. As documented by the literature, FAI salt can remove OA oleylammonium ligands by cation exchange and MeOAc can hydrolyze to acetic acid and methanol to remove oleic acid ligand from the surface.¹⁰² It should be noted that the hydrolysis of MeOAc without the aid of acids and bases is slow. Therefore, a mild process for controlling the density of surface ligands can be achieved by using a secondary amine, such as di-*n*-propylamine (DPA). It is reported that by treating CsPbI_3 PQDs with DPA, it is possible to promote efficient ligand removal and maintain the integrity of the devices at the same time.¹⁸⁰

Likewise, it is possible to enhance device stability through the rational design of devices. The spiro-OMeTAD HTL is one of the major factors contributing to the degradation of perovskite PVs, while polymeric HTMs are intrinsically more stable compared to small organics. It is also possible to prevent the degradation of cells using PTB7 as a promising substitute to spiro-OMeTAD.¹⁵⁹ In this regard, Yang et al.¹⁸¹ reported the unencapsulated polymer-modified devices by depositing polythiophene on the top of CsPbI_2Br QDs with great stability maintained around 90% of their initial PCE after 960 h of storage in a dry glovebox.

Ultimately, the fabrication of SC devices can also have a direct impact on their operational stability. The layer-by-layer deposition could deteriorate the stability due to the washing of each layer and the resulting defects in the QD film. As a matter of fact, it is highly recommended that QD layers be deposited in sequence.

3.1.3. Summary. Altogether, in this review, the PQD function in SCs and the recent advances of emerging PQDSCs were summarized from the viewpoints of device structure and improving performance. In this regard, we outlined the structure and chemical interface engineering of PQDSC, surface ligand management, surface passivation and post-treatment of PQDs, and their charge separation management and stability improvement. Through the research avenue, it was recognized that the addition of volatile organic cations into PQDs would raise concerns about thermal stability during long-term operational conditions. Consequently, improving the efficiency of all-inorganic PQDSCs remains a challenge. However, thanks to extensive research efforts, the PQDSCs' efficiency has been boosted from 10.8% in 2016 to over 16% in 2020. Based on this success, the authorized efficiency for PQDSCs is now comparable with organic SCs and other thin-film technologies, transforming PQDSCs into an emerging candidate for the next-generation SCs.

3.1.4. Future Perspectives. While colloidal PQDs exhibit significantly enhanced phase stability at room temperature in comparison with their bulk counterparts, the stability of PV devices based thereon under operation has still emerged as the most urgent problem confronting the perovskite research community. Scientists have invested considerable efforts in enhancing the stability of PQDSCs that have been focused

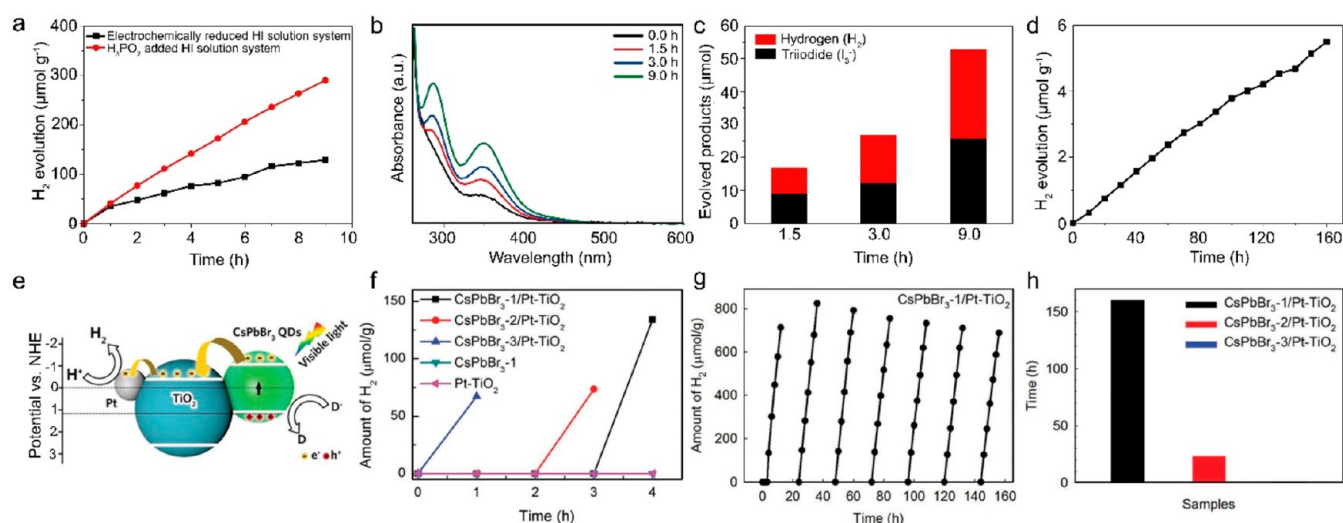


Figure 11. (a) Photocatalytic H₂ evolution from the MAPbI₃ powder in a saturated solution. The aqueous HI solutions were prepared by electrochemical reduction (black squares) and H₃PO₂ addition (red circles), respectively. (b) Photocatalytic reaction time-dependent UV–vis spectra of diluted saturated solutions for I₃⁻ titration. The saturated solution was made by adding MAPbI₃ powder to the electrochemically reduced HI solution. (c) Quantitative comparison between the evolved H₂ and I₃⁻. (d) Stable photocatalytic H₂ evolution produced by the MAPbI₃ powder in the saturated solution for 160 h. H₃PO₂ was added to the HI solution. Reproduced with permission.¹⁸⁵ Copyright 2016, Nature. (e) Illustration of the photocatalytic activity on the CsPbBr₃ QDs/Pt-TiO₂ composite for H₂ production. (f) Photocatalytic H₂ evolution activity of various samples during the first several hours (visible light $\lambda > 420$ nm). (g) Long-term H₂ generation of CsPbBr₃-1/Pt-TiO₂ photocatalyst under visible light irradiation. (h) Duration of photocatalytic H₂ evolution reaction for CsPbBr₃-1/Pt-TiO₂, CsPbBr₃-2/Pt-TiO₂, and CsPbBr₃-3/Pt-TiO₂. Reproduced with permission.¹⁸³ Copyright 2019, Wiley.

primarily on three common fundamental strategies. The first strategy focuses on compositional engineering and/or doping of PQDs, the second one is based on the surface treatment of QD film and the third strategy relies on optimization of SC device structure (ETL, active layer, and HTL). However, an extensive investigation is still necessary to address many open questions to fully understand their impact on next-generation applications. What is more, although CsPbI₃ PQDSCs have demonstrated a high V_{OC} in comparison to thin-film PSCs in the same bandgap energy, they still suffer from a poor J_{SC} due to surface defects and insulating organic ligands between PQD interfaces. Hence, the issue of low carrier mobility in PQD films should be addressed to achieve high-efficiency PQDSCs. Recently, Han and colleagues¹¹⁶ reported a novel strategy in which utilizing ionic liquids with high polarity and electron-donating character could boost the mobility of PQD films in PQDSCs resulting in a significant increase in efficiency. Chen et al.¹⁸² also reported a “surface surgery treatment” of PQDs for resurfacing PQDs by exploiting multidentate ligands, in which the surface defect passivation of PQDs by occupying I⁻ vacancies and cross-link PQDs with multidentate ligands as a charger bridge could substantially boost the efficiency. The incidence of these concerns will eventually lead to a greater emphasis on finding other novel pathways to improve PQD device efficiency and stability through enhanced optoelectronic properties.

3.2. PQDs in Photocatalysis Application. Photocatalysis introduces one of the most beneficial approaches to tackling the controversial concerns related to the energy crisis and environmental issues.¹⁸³ Over the past few years, halide perovskites by showing peculiar characteristics in the field of photovoltaics, have been included in the photocatalysis library and thus, displayed huge potential for energy conversion through harnessing solar power. Some of the exceptional features which make these materials favorable for photo-

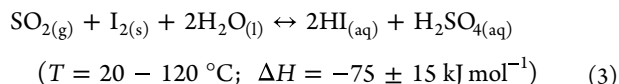
catalysis are as follows, superior light-absorbing factors, small binding energies of the exciton, long lifespan of charge transporter, and diffusion lengths ($\approx 1 \mu\text{m}$).¹⁸³

Additionally, LHP-based CQDs fabricated by the hot injection method¹⁸⁴ containing neat size and shape may hold further merit as opposed to those of LHP NC with larger size and irregular shapes in photocatalysis, three of which are further elaborated. First of all, they intrinsically enjoy a larger surface area, and the energy levels can be adjusted in terms of size¹⁸⁴ to regulate the electronic transmission systems with the substrates. Second of all, CQDs can create a homogeneous colloidal solution in many organic solvents regarding their high dispersibility property, while maintaining their heterogeneous-like catalytic traits including recyclability and separability. Last but not least, surface-controlled catalysis with particular specifications for different substrates can be achieved owing to their monodisperse, fine-tuned size, and morphological configuration of the surface.¹⁸⁴ Recently, several reports have been released in which the diverse attributes of these perovskites have been exploited for: photocatalytic hydrogen generation, degradation of organic compounds and antibiotic residues in organic systems, CO₂ reduction, stereoselective C–C oxidative coupling, polymerization reactions, NO removal, and organic transformations, each of which is briefly discussed hereunder.

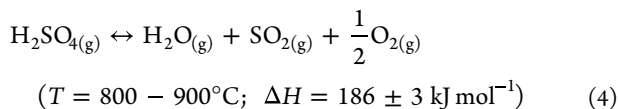
3.2.1. Photocatalytic Hydrogen Generation. Hydrogen can be produced via multiple approaches such as photo(electro)-catalytic water splitting, and thermochemical and biological routes.^{186–189} The photocatalysis through lead halide perovskite materials came into the picture only after Park et al.¹⁸⁵ demonstrated the photocatalytic hydrogen generation from hydriodic acid using methylammonium lead iodide for the first time in 2016. By this time, there had been enormous research on the thermochemical water splitting of HI to produce H₂.^{190–192} Yet, hydrogen production on a large scale is still a

formidable challenge. One promising technology for mass-scale hydrogen production is the sulfur–iodine cycle, which has been used by researchers over the past few decades.¹⁹³ This cycle has high efficiency of around 50% and this can be further improved by optimizing the parameters involved in the sulfur–iodine process.^{194,195} It consists of the following reactions:^{196,197}

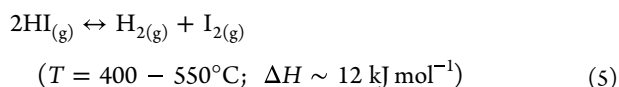
Bunsen reaction:



Sulfuric-acid decomposition:



Hydrogen-iodide decomposition:



Net reaction:



In eq 3 (Bunsen reaction), the reaction of sulfur dioxide, iodine, and water produces a mixture of hydrogen iodide and sulfuric acid. After the separation of hydrogen iodide and sulfuric acid, the two products are decomposed further in eq 4 (sulfuric-acid decomposition) and (5) (hydrogen-iodide decomposition). Equation 4 generates water, sulfur dioxide, and oxygen, whereas eq 5 produces hydrogen and iodine. The products sulfur dioxide and water from eq 4 and iodine from eq 3 are recycled back to the Bunsen reaction and this process goes on. The net sum of all the reactions involved in this cycle is the splitting of water into hydrogen and oxygen in a carbon dioxide-free environment.

In addition to this process, the solar-induced splitting of hydrohalic acids (HX) is another favorable alternative for conventional water splitting. For the splitting of HX which involves two electrons, a smaller overpotential is demanded compared to water splitting with four electrons.^{198,199} It can be concluded that the overpotential for iodide ion oxidation is close to zero.^{200,201} The key challenge in materials design for the photoinduced HI splitting originates from the strong acidity of aqueous HI, implying that most materials are not able to tolerate acidic conditions.²⁰² Considering the above-written points, Park et al.¹⁸⁵ prepared methylammonium lead iodide (MAPbI₃) to be utilized for hydrogen generation via HI splitting.

The photocatalytic reaction proceeded by irradiation of visible light ($\lambda \geq 475 \text{ nm}$) (Figure 11a) over precipitated MAPbI₃ powder as the photocatalyst to dissociate HI into H₂ and I₃[−]. To evaluate the role of visible light and a photocatalyst, the reaction was conducted in the absence of light and a catalyst. It was observed that no H₂ gas was identified in the absence of each of these parameters, signifying that the generated H₂ gas can be solely correlated to MAPbI₃ powder. The presence of I₃[−] ions was proved by the UV–vis spectra (Figure 11b). The quality check of the products confirmed that overall HI splitting took place with no side reactions (Figure 11c). Over time, the evolved hydrogen rate

dwindled as a consequence of interference caused by the light absorption of I₃[−]. To overcome this problem, the authors selected H₃PO₂—known as a good reducing agent for I₃[−]—to be added to the saturated solution to reduce I₃[−] to I[−]. Based upon the results, Park et al.¹⁸⁵ claimed that MAPbI₃ was maintained photocatalytically active after 160 h of continuous illumination, as depicted in Figure 11d. The remarkable durability was ascribed to the quasi-stable property of the dynamic equilibrium between MAPbI₃ powder and the saturated solution.

Following this study, an explosion of research on PQD-based photocatalysis occurred. Undoubtedly the photocatalyst is the crucial ingredient for photocatalysis. Yet, insufficient tolerance of ionic compounds in the vicinity of moisture or air is a formidable challenge that restricts the rapid advance in photocatalysis.^{203,204} Due to the far-reaching impact of the photocatalysts in various areas, the development of tolerant photocatalysts able to function under ambient conditions cannot be ignored. As mentioned, in the case of PQDs, one of the options is through modifying the surface with surface ligands that so far worked well for optoelectronic devices.^{6,205} For the light-driven photocatalytic hydrogen evolution, however, Xiao et al.¹⁸³ took advantage of this strategy for the first time to stabilize CsPbBr₃ QDs against destruction in moisture while sustaining efficient charge transfer for redox reactions.

Xiao et al.¹⁸³ synthesized CsPbBr₃ QDs with the aid of a hot-injection method, which on the surface were protected by OLA and OA as surface ligands (Figure 11e). Samples were labeled CsPbBr₃-1, CsPbBr₃-2, and CsPbBr₃-3 indicating the CsPbBr₃ QDs rinsing one to three times, individually. To improve the photocatalytic performance, a heterojunction was developed between CsPbBr₃ QDs and TiO₂ (anatase) nanoparticles (NPs) with which charge separation could be promoted.^{206,207} Prior to this combination, TiO₂ was decorated with Pt nanoparticles as the cocatalyst by the previously reported method.²⁰⁸ In addition, the ratio of CsPbBr₃ QDs/Pt-TiO₂ were selected to be 9.86, 9.10, and 8.63 wt % for CsPbBr₃-1, CsPbBr₃-2, and CsPbBr₃-3, respectively. Finally, CsPbBr₃ QDs/Pt-TiO₂ photocatalysts were exploited in a gas/solid photocatalytic system to generate hydrogen (Figure 11e).

The authors performed control experiments to realize the mechanism of the reaction. With only methanol, the rate of generated H₂ was merely 20% of the solution of water/methanol, indicating that water was an essential factor in completely oxidizing the methanol under anaerobic conditions.²⁰⁹ Nonetheless, for the reaction with only water as the solution, no H₂ generation was reported, emphasizing the high barrier for water splitting under light illumination. Furthermore, the photocatalytic reactions conducted on the composites substantiated that these materials were active under visible-light illumination ($\lambda > 420 \text{ nm}$) (Figure 11f). In addition, control experiments showed that in the presence of neither light nor catalyst, no H₂ was detected. Moreover, no H₂ was produced using Pt-TiO₂ photocatalyst or CsPbBr₃ QDs individually within 5 h. This behavior was justified by the following facts; first, TiO₂ is active under the UV region, and second, as an individual photocatalyst CsPbBr₃ QDs are not competent enough to drive H₂ from water.

From photocatalytic results illustrated in Figure 11f, CsPbBr₃-1/Pt-TiO₂ and CsPbBr₃-2/Pt-TiO₂ required 2–3 h to be activated under visible light. In contrast, CsPbBr₃-3/Pt-

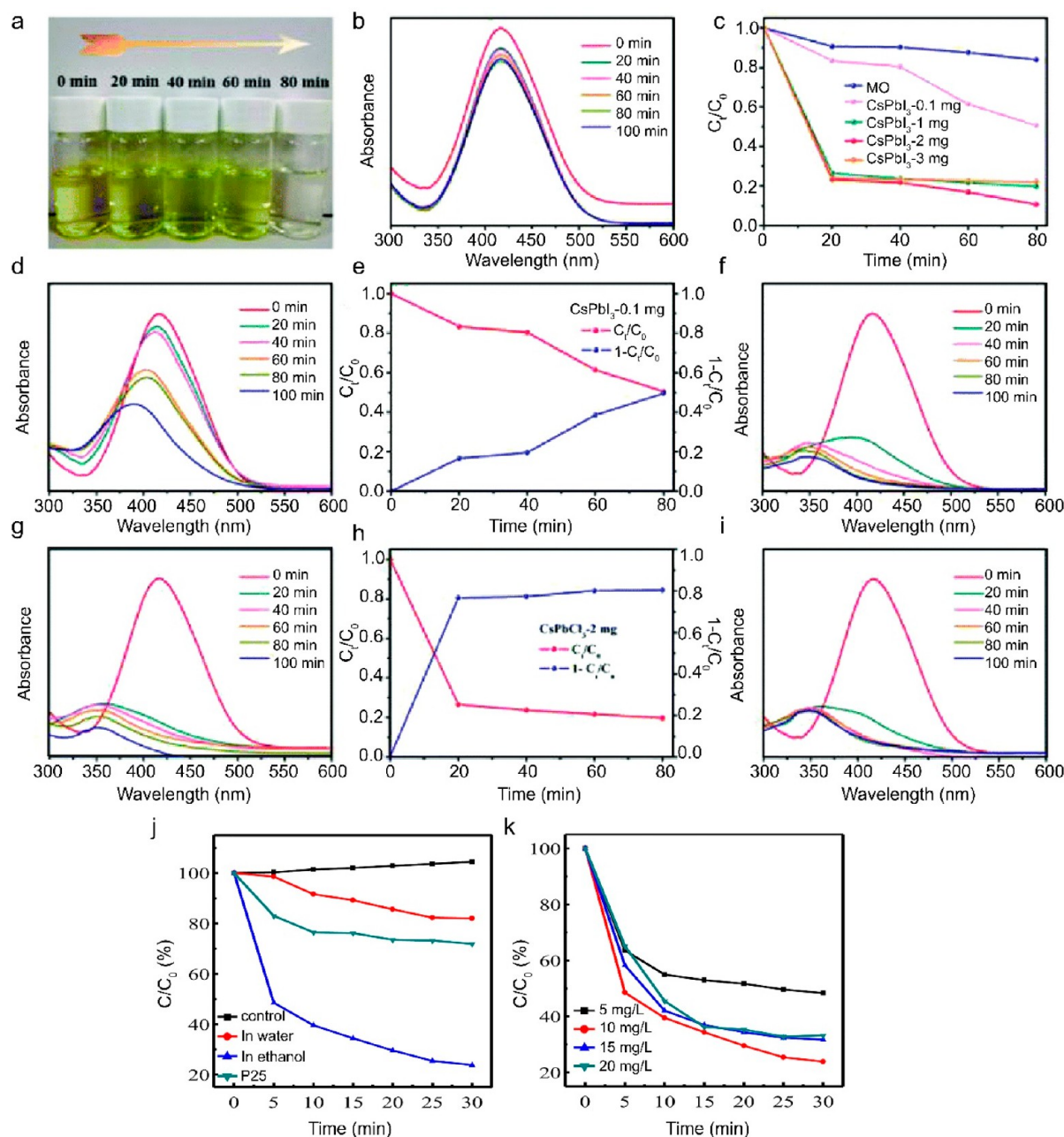


Figure 12. (a) Image of MO degradation using CsPbCl₃ QDs. (b) UV–vis spectra of MO degradation in the absence of catalysts. (c) Concentration (C_t/C_0) changes of MO with different catalysts under visible-light irradiation. Panels d, f, g, and i show the UV–vis spectra of MO degradation using CsPbCl₃-0.1 mg, CsPbCl₃-1 mg, CsPbCl₃-2 mg, and CsPbCl₃-3 mg, respectively. Changes in the MO degradation efficiency ($1 - C_t/C_0$) using (e) CsPbCl₃-0.1 mg and (h) CsPbCl₃-2 mg. Reproduced with permission.²¹¹ Copyright 2017, The Royal Society of Chemistry. (j) Concentration (C_t/C_0) changes of TC-HCl without CsPbBr₃ (Control experiment), with CsPbBr₃ in water, with CsPbBr₃ in ethanol, and with P25 in ethanol, respectively. (k) Photodegradation of TC-HCl at different concentrations on CsPbBr₃ in ethanol. Reproduced with permission.²¹⁴ Copyright 2020, Elsevier.

TiO₂ demonstrated H₂ generation within first hour of light irradiation. Besides, in terms of stability tests (refer to Figure 11g,h), the following trend was found CsPbBr₃-1/Pt-TiO₂ (160 h) < CsPbBr₃-2/Pt-TiO₂ (≤23 h) < CsPbBr₃-3/Pt-TiO₂ (≤2 h).

3.2.2. Photocatalytic Degradation of Organic Compounds. Man-made environmental contamination has turned into a consistent challenge with the galloping rate of population expansion and economic growth. More to the point, the released sewage from industrial plants usually contains dyes, metal ion-containing compounds, antibiotics

remaining, harmful microorganisms, and other strong contaminants, which severely threaten the safety and security of the ecosystems.²¹⁰ Needless to mention that the transformation of organic dyes into benign materials is critical for human life and eco-friendly growth.²¹¹ In addition to this, antibiotic residues, one of the medicine contaminants, leave pernicious effects on human health and the ecosystem, which have sparked a great deal of interest. Tetracycline hydrochloride (TC-HCl) for example is one of the most widespread antibiotics, and its residues inflict substantial harm on the ecological environment. Considering the above-written con-

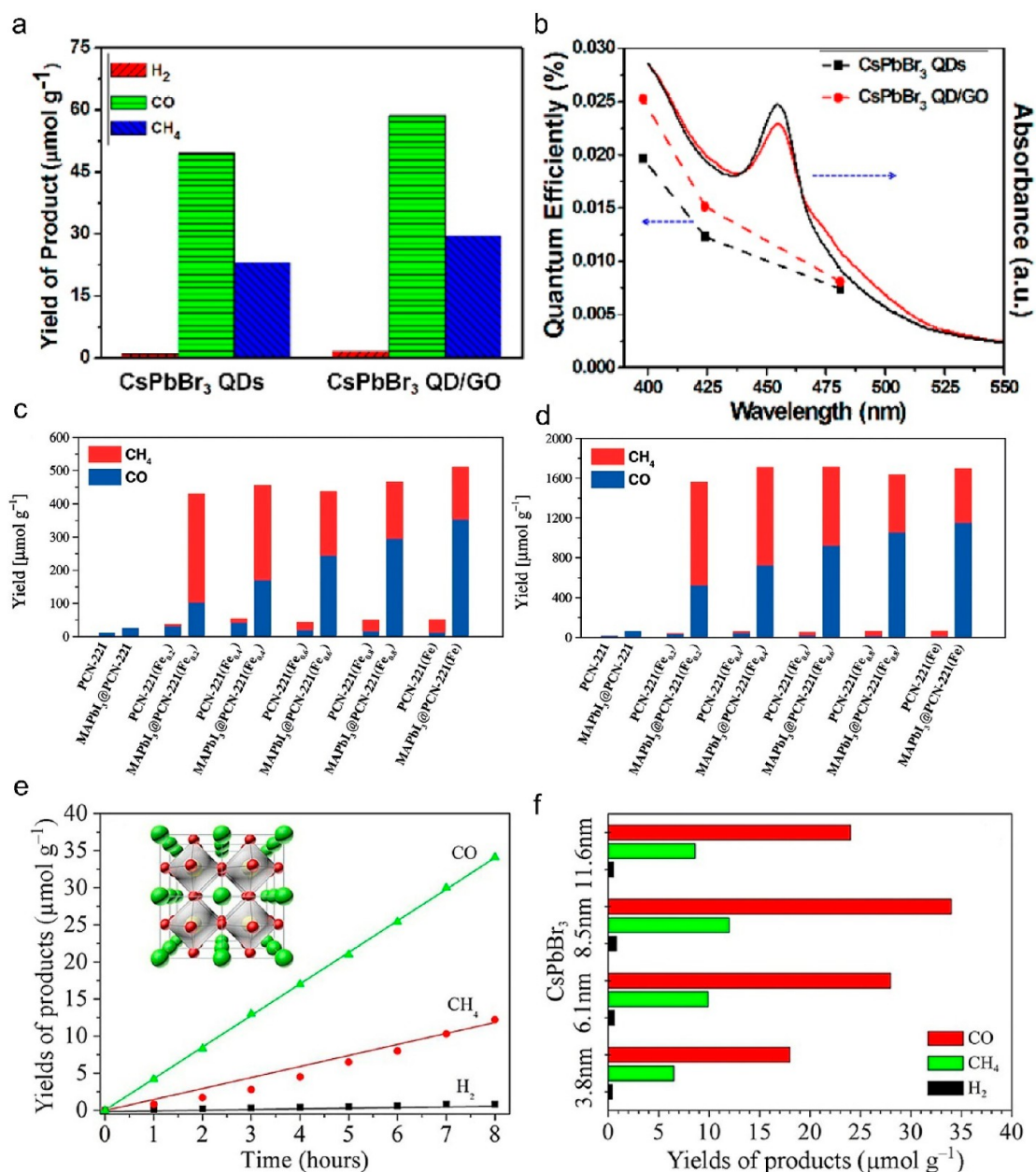


Figure 13. (a) Photocatalytic performance: yield of the CO_2 reduction products after 12 h of photochemical reaction. (b) UV–vis absorption spectra and the EQE plots. Reproduced with permission.²²³ Copyright 2017, The American Chemical Society. The yields for CO_2 reduction to CH_4 and CO with $\text{PCN-221}(\text{Fe}_x)$ and $\text{MAPbI}_3@ \text{PCN-221}(\text{Fe}_x)$ as photocatalysts in the CO_2 -saturated ethyl acetate/water solution after (c) 25 h, (d) 80 h of irradiation under 300 W Xe lamp, with the light intensity of 100 mW cm^{-2} . Reproduced with permission.²²⁰ Copyright 2019, Wiley. Solar CO_2 reduction into fuels under 300 W Xe lamp irradiation for CsPbBr_3 QDs, (e) 8.5 nm CsPbBr_3 QDs, and (f) tunable CsPbBr_3 QDs with different particle sizes. Reproduced with permission.²²⁴ Copyright 2017, Wiley.

cerns, it is of significant importance to transform the detrimental residues into nontoxic and innocuous contents for the sake of protection of the environment and long-term development.^{212,213}

Several methods including catalytic oxidation, electrolyzing, and adsorption via activated carbon have been employed to remove antibiotics remaining from water. Due to some flaws correlated with each mentioned method, photocatalysis has arisen as one of the research hotspots thanks to its high performance; without any peripheral contaminations as a result of the degradation of antibiotics remaining. Therefore, enormous efforts have been put forward to craft and harness

visible-light-driven photocatalysts based on perovskite quantum dots with suitable traits to take the most advantage of solar energy.²¹⁰ To this end, Gao et al.²¹¹ released a report on the photocatalytic degradation of methylene orange (MO) for the first time with the aid of CsPbX_3 ($\text{X} = \text{Cl}, \text{Br}, \text{and I}$) QDs. The CsPbX_3 QDs were prepared by an emulsion and demulsion method at room temperature. Thereafter, the photocatalytic performance of as-obtained QDs was examined through decomposition of an ordinary organic pollutant. CsPbX_3 QDs were added in dissimilar quantities: 0.1 mg, 0.5 mg, 1 mg, 1.5 mg, 2 mg, and 3 mg. These samples were denoted as

CsPbX₃-0.1 mg, CsPbX₃-0.5 mg, CsPbX₃-1 mg, CsPbX₃-1.5 mg, CsPbX₃-2 mg, and CsPbX₃-3 mg.

Under visible-light illumination, merely 17% of MO degradation was achieved after 80 min without the presence of the catalyst (see Figure 12a–c). In sharp contrast, utilizing the CsPbCl₃ as the photocatalyst, the experiments showed a direct relationship between the efficiency of photocatalytic degradation of MO and CsPbCl₃ content. The concentration (C_t/C_0) changes of MO with various photocatalysts can be seen in Figure 12c. In the absence of the catalyst, the photodegradation efficiency is insignificant. The same fashion can be followed in the UV–vis absorption spectra. The conspicuous photocatalytic performance can be discerned following the addition of CsPbX₃ QDs (Figure 12d,f,g,i).²¹¹ Finally, the MO degradation efficiency ($1 - C_t/C_0$) was conducted over two catalysts, namely, CsPbCl₃-0.1 mg and CsPbCl₃-2 mg, and the outcomes were summarized in Figure 12a,h.

Currently, the majority of photocatalysts are designed to eliminate pollutants in aqueous solutions in preference to nonaqueous systems.^{215,216} Nonetheless, some detrimental agents such as tetrachloroethylene and trichloroethylene in chemical and pharmaceutical applications can be dispersed via utilizing organic systems.^{217,218} To preclude the widespread diffusion of these hazardous chemicals into the soil or water, practical degradation approaches are needed for the treatment of nonaqueous solutions.²¹⁷ For the sole sake of this purpose, Qian et al.²¹⁴ prepared CsPbBr₃ QDs by antisolvent precipitation and subsequently applied for the photocatalytic degradation of both tetracycline hydrochloride (TC-HCl) and MO in ethanol solution.

From Figure 12j for the case of the control experiment, no apparent decrease in TC-HCl concentration was detected in the absence of the catalyst; in fact, within the running period, TC-HCl concentration negligibly increased, which was attributed to the evaporation of ethanol during the test process. Moreover, only 18% of TC-HCl was decomposed after the addition of prepared CsPbBr₃ into the water solution of TC-HCl. In contrast, once ethanol was used as the solvent, photocatalytic activity was tremendously promoted, and consequently, the TC-HCl removal reached 76% within 30 min under visible-light exposure. In addition, the photodegradation of TC-HCl with varied concentrations including 5, 10, 15, and 20 mg/L was realized maintaining identical reaction conditions, individually, as portrayed in Figure 12k. On the basis of the findings, the best degradation performance on TC-HCl belonged to the catalyst in alcohol at the TC-HCl concentration of 10 mg/L. These results demonstrated that CsPbBr₃ QDs own remarkable photocatalytic activity once they are exposed to visible light.

Even though PQD-based photocatalysts have exhibited tremendous potential for the deterioration of organic dyes and antibacterial species, they sometimes are constrained by the immediate recombination of inner paired-up electron–hole. To circumvent this bottleneck, the combination of individual PQDs with another promising semiconducting material to construct a composite could be a possible solution reported.²¹⁰ The combination of AgBr/Ag with PbBiO₃Br showed an improvement in terms of performance and stability.

3.2.3. Visible-Light-Driven Photocatalytic CO₂ Reduction. The rapid depletion of fossil fuel resources (coal, oil, natural gas), and inordinate emission of carbon dioxide (CO₂) have urged the scientific community to seek viable clean energy and

in turn, dwindle the greenhouse effect in current society.²¹⁹ Direct transformation of CO₂ to value-added chemicals under visible light over robust catalysts is one of the attractive scenarios to prevent both depleted fossil fuels and global warming generated by greenhouse gas emissions.^{220–222} For this purpose, LHP materials have received considerable value in photovoltaics and optoelectronic applications by virtue of their distinctive optical attributes.²²¹ Yet, the inorganic halide PQDs such as CsPbBr₃, being remarkable materials for PSCs and light-emitting diode devices, have poorly been studied for photochemical CO₂ conversion to chemicals following their fragile stability in the vicinity of wetness or polar solvent.²¹⁹

Keeping in mind the above-written merits of CsPbBr₃, its catalytic performance is still under question by the serious charge recombination. Thus, far, a wide variety of cocatalysts have been applied to integrate with CsPbBr₃ to form composites, such as rGO, g-C₃N₄, and Pd nanosheets, metal–organic frameworks (MOFs), and MXene nanosheets.²²² The typical example that we can name is Xu et al.²²³ who manufactured CsPbBr₃ QDs and a CsPbBr₃ QD/graphene oxide (CsPbBr₃ QD/GO) composite for the photocatalytic reduction of CO₂ to EtOAc. In this work, the photocatalytic CO₂ reduction measurements were carried out in a sealed Pyrex bottle with EtOAc as the solvent. The photocatalytic CO₂ reduction to produce CH₄ and H₂ was evaluated after 12 h, and the results are summarized in Figure 13a,b and Table 2. Xu et al.²²³ further claimed that there was a

Table 2. Summary of the Photocatalytic CO₂ Reduction Performances after 12 h of Constant Illumination.²²³

Sample	R(CO) ($\mu\text{mol g}^{-1}$)	R(CH ₄) ($\mu\text{mol g}^{-1}$)	R(H ₂) ($\mu\text{mol g}^{-1}$)	R_{electron}^a ($\mu\text{mol g}^{-1}$)	select. for CO ₂ red. ^b (%)
CsPbBr ₃ QDs	49.5	22.9	1.07	284.7	99.3
CsPbBr ₃ QD/GO	58.7	29.6	1.58	357.4	99.1

^a R_{electron} is the rate of electron consumption for the reduced product; $R_{\text{electron}} = 2R(\text{CO}) + 8R(\text{CH}_4) + 2R(\text{H}_2)$. ^bSelectivity for CO₂ reduction = $[2R(\text{CO}) + 8R(\text{CH}_4)]/R_{\text{electron}} \times 100\%$.

linear relationship between the concentrations of all the products and time. In this sense, the averaged electron consumption (R_{electron}) for the CsPbBr₃ QDs was 23.7 $\mu\text{mol/g h}$ in the course of the reaction, and a R_{electron} of 29.8 $\mu\text{mol/g h}$ was achieved for the CsPbBr₃ QD/GO composite. In light of the findings, the photocatalysis of synthesized QDs in this study—at the time—surpassed those of CdS and other newly manufactured photocatalysts under visible-light irradiation.

As mentioned, the combination of LHPQDs and graphene oxide²²³ or g-C₃N₄²²¹ leads to efficient charge separation, bringing about enhanced catalytic activity toward CO₂ reduction. Nonetheless, the poor stability of LHPQDs and the scarcity of active catalytic sites hinder their performance in photocatalytic CO₂ reduction. To tackle these challenges, Wu et al.²²⁰ came up with a brand new solution, which was to wrap LHPQDs of MAPbI₃ in the pores of Fe-porphyrin-based MOFs of PCN-221(Fe_x) to manufacture a series of MAPbI₃@PCN-221(Fe_x) composites as photocatalysts for visible-light-induced CO₂ reduction ($x = 0–1$). It is postulated that MOFs hold promising intrinsic activities for the conversion of CO₂.^{225–229} Besides their interesting catalytic properties, the

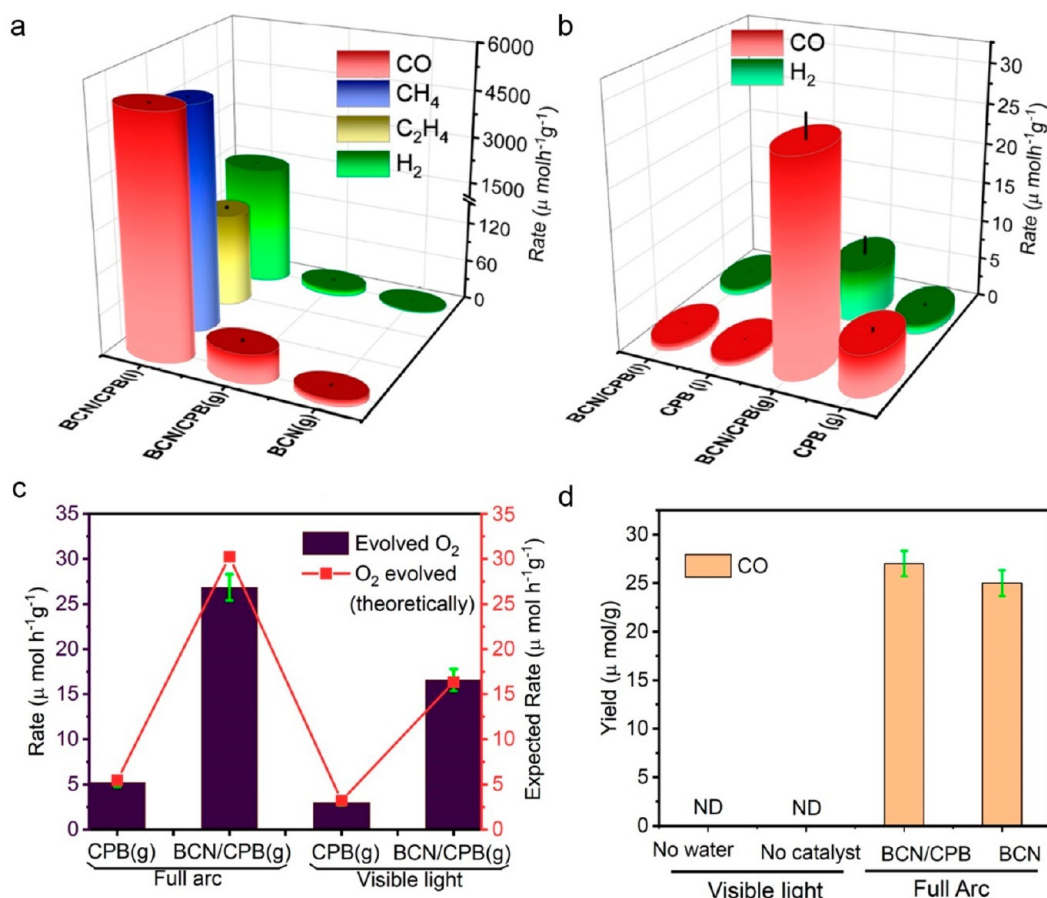


Figure 14. Rate of product formation with BCN/CsPbBr₃ composite catalyst in different reaction conditions. A photocatalytic study was done under (a) UV–vis light ($\lambda > 100$ nm) and (b) visible light ($\lambda > 400$ nm). CPB, CsPbBr₃; BCN, boron carbon nitride. The “g” and “l” refer to the reaction condition. Catalysis was done at catalyst–gas interface conditions (g) and catalyst–liquid interface conditions (l). The rate is shown on a logarithmic scale. (c) Rate of O₂ evolution with respect to theoretically calculated O₂ evolution rate for CPB and BCN/CPB composite catalyst in gas phase condition. (d) Control experiments with BCN and BCN/CPB catalyst in absence of CO₂ and water with 5 mg of catalyst. Experiments were also done in the absence of water vapor and in the presence of CO₂ in visible light. Another control study was done in the absence of a catalyst but in presence of CO₂ and water vapor. ND: Not detected. Reproduced with permission.²⁴⁴ Copyright 2021, The American Chemical Society.

framework of MOFs can elevate the longevity of LHP QDs.^{230,231} The close attachment of LHPQDs to the catalytic active spots in MOFs can minimize the charge-transfer distance, thus promoting the photoinduced charge separation efficiency of LHPQDs and catalytic performance of MOFs. To fulfill this objective, Wu et al.²²⁰ conducted a consecutive deposition technique to wrap the MAPbI₃ QDs within the pores of PCN-221(Fe_x), as illustrated in Figure 13c,d. The photoreduction experiments of CO₂ were carried out in a CO₂-saturated EtOAc solution containing a small amount of water. As depicted in Figure 13c, concerning the bare samples of PCN-221(Fe_x), CO₂ reduction increased with an increase in Fe amount owing to the increase of Fe catalytic sites. With reference to the results, among PCN-221(Fe_x) catalysts, the most efficient one generated only 13 and 38 mmol g⁻¹ of CO and CH₄, respectively, after 25 h of illumination. After that time, the PCN-221(Fe_x) photocatalysts were demolished and lost their catalytic activity.

Under the same exposure period of 25 h, Fe-free MAPbI₃@PCN-221 likewise displayed negligible photocatalytic performance toward CO₂ reduction (refer to Figure 13c). On the contrary, the composite photocatalysts containing both MAPbI₃ QDs and Fe demonstrated substantial enhancement of photocatalytic activity for CO₂ reduction, bringing about

notably improved yields for both CO and CH₄ and thereby confirming the indispensable role of Fe as a catalytic site for photocatalytic CO₂ reduction. Finally, this work proved that the as-obtained composites of MAPbI₃@PCN-221(Fe_x) possess greater durability compared with other LHPQDs-based photocatalysts for CO₂ reduction.^{221,223,224,232}

In another study, Hou et al.²²⁴ did survey the in-depth size-dependency of the photocatalytic CO₂ reduction system at the atomic level through a governable colloidal approach. Not to mention that CsPbBr₃ QDs, besides possessing a suitable bandgap, are favorable semiconducting materials with a high capacity to absorb sunlight.^{13,14,233,234} In addition, the optical band gaps and PL spectra are adjustable over the whole visible region owing to quantum size effects. Moreover, the prolonged carrier lifespan which can be uncovered by time-resolved photoluminescence (TRPL) elucidates the electron–hole separation efficiency. Consequently, the CsPbBr₃ QDs bearing an optimum size yield the highest photochemical conversion efficiency for the solar CO₂ reduction reaction.

Keeping in mind the above-written advantages, Hou et al.²²⁴ developed CsPbBr₃ QDs adopting the solution phase synthesis technique. To put the as-prepared CsPbBr₃ QDs into practice, Hou et al.²²⁴ utilized them as photocatalysts for the photocatalytic CO₂ reduction reaction. Following the photo-

catalysis, the major products (CO , CH_4 , and H_2). There was a linear growth between all the product generation and the prolonged time. The averaged electron yield (R_{electron}) rate for the CsPbBr_3 QDs— $20.9 \mu\text{mol g}^{-1} \text{h}^{-1}$ during the entire reaction—was estimated based on the following equation: $R_{\text{electron}} = 2R_{\text{CO}} + 8R_{\text{CH}_4} + 2R_{\text{H}_2}$. Such a high product formation rate of the CsPbBr_3 QDs was much larger than that of previously reported WO_3 , CdS , Cu_2S , and PbS/TiO_2 , demonstrating the superior catalytic activity of CsPbBr_3 QDs.^{235–237}

Needless to mention that colloidal PQDs are attractive candidates for the photoreduction of CO_2 on account of their abilities to modulate their catalytic features through size modifications.^{13,14,233,234,238} To build a logical connection between the quantum size effect and solar-induced CO_2 size-controllable CsPbBr_3 QDs with different sizes were employed as photocatalysts for the CO_2 reduction experiments. From Figure 13f, the highest yields of CO , CH_4 , and H_2 in the span of 8 h under solar light exposure were obtained for the 8.5 nm CsPbBr_3 QDs. The CsPbBr_3 QDs with a size of 8.5 nm enjoy a strong optical absorption capacity, high surface area, and large active sites, all of which encourage the faster charge transfer to the surface, shorten the charge transference path, and at the same time impede the electron–hole recombination. As a whole, while downsizing brings about extreme aggregation, the large size diminishes the surface area,^{13,14,233,234,238} regulating the optical absorption and charge transfer, and separation.²³⁹ In a conclusion, the engineering of the particle size (refer to Figure 13e,f) is of paramount significance to promote solar CO_2 reduction.

Considering the huge attention given to this research, water has been selected as the prime candidate for the conversion of CO_2 .²⁴⁰ Nevertheless, the overall efficiency of this process is highly dependent on the solubility of CO_2 in water, which is low and thus not satisfactory.²⁴¹ To overcome this problem, people have been using other solvents such as ACN, EtOAc, aqueous carbonate solutions, and aqueous NaOH solutions along with several hole scavengers like triethanol amine (TEOA), triethyl amine (TEA), etc.^{223,242,243} Being said this, some solvents can be photoactive under light illumination and converted to one of the products of CO_2 reduction, resulting in faulty conclusions on the efficiency of the catalyst. This fact is often ignored when selecting the appropriate solvent.

Das et al.²⁴⁴ performed a thorough investigation to determine the effect of the solvent on the photocatalytic CO_2 reduction performance over a halide perovskite catalyst BCN (boron carbon nitride)/ CsPbBr_3 in the presence of hole scavengers under artificial UV–vis light and visible light. Das and colleagues illustrated the rate of products under various reaction conditions in Figure 14. From this figure, pristine CsPbBr_3 (CPB) and the BCN/ CsPbBr_3 composite (BCN/CPB) displayed CO and H_2 production under full arc (200–2400 nm) in vapor phase conditions. In contrast, in liquid phase conditions, the composite catalyst generated a very high amount of CO , CH_4 , C_2H_4 , and H_2 (Figure 14a), which could be ascribed to ACN photolysis. As depicted in Figure 14b both CPB and BCN/CPB produced CO and H_2 under visible light in vapor phase conditions. Interestingly, in the liquid phase condition (10 mL of ACN and 40 μL of water) the photocatalytic activity was negligible. The authors related this phenomenon to poor water adsorption on the catalyst surface due to an insufficient supply of water in the medium.

The authors further determined the quantity of oxidation product (O_2) to validate whether the products developed from the photocatalytic redox process or solvent photolysis. In gas medium BCN/CPB or CPB both catalysts under visible light showed a mass balance between reduced products (CO and H_2) and oxidized product (O_2) (Figure 14c). However, some amount of CO was generated from photodegradation of BCN under full arc for BCN/CPB and only BCN in the absence of CO_2 (Figure 14d), which might be the reason for a slightly lower O_2 evolution rate compared to the expected value for BCN/CPB(g) (Figure 14c).

On the other side, for the liquid phase reaction, O_2 was not produced in the presence of CPB and BCN/CPB catalysts under the full arc or visible region, demonstrating the fact that the obtained carbonaceous products in the ACN/water mixture under full arc did not originate from CO_2 reduction. The produced CO originating from the reduction of CO_2 by CPB/BCN composite under visible light in vapor phase conditions was further confirmed using $^{13}\text{CO}_2$. Furthermore, no gas formation was detected under visible light in the absence of water vapor or the absence of catalyst, which confirmed that CO_2 to CO production was just not a stoichiometric reaction where the O of CO_2 was left behind in the solid material. The observed CO came from the photocatalytic process where water oxidation to O_2 happened by photogenerated holes.

The control experiments in this study verified the photolysis of CH_3CN , EtOAc, TEA, and TEOA under UV–vis light even in the absence of any catalyst is plausible to produce CO , CH_4 , C_2H_4 , and H_2 . The selection of the right solvent for photocatalysis is critical as if it is overlooked, it can lead to overestimation of catalytic efficiency and product selectivity.

3.2.4. Stereoselective C–C Oxidative Coupling Reactions. Colloidal QDs as photocatalytic materials demonstrate a few distinctive benefits in contrast to metal-based coordination complexes and organic dyes. First, colloidal QDs can be simultaneously utilized as both photosensitizers and catalysts, outperforming common photocatalytic approaches in which photosensitizers and catalysts are usually separate components.²⁴⁵ Second, frequently used precious metal-containing complex-based catalysts impose a pricy procedure which is not the case in typical QDs (CdS , CdSe , InP QDs, or perovskite QDs). Third, the modulation of surface ligands of the QDs can be obtained without affecting the redox properties or the absorption characteristics of the QDs, compared to complex-based materials in which electronic structure can be significantly tuned by ligand species. Lastly, the redox potentials of the QDs, which are dependent on their bandgaps, can be modified for diverse catalysis by means of engineering the size, morphology, and/or composition of the QDs.²⁴⁵

It was only recently that we have begun to comprehend the photocatalytic reactions with stereoselectivity over PQDs. Despite their superb qualities in phytochemistry and optoelectronics, additional investigations are required to make the most of their photocatalytic potential. In this respect, Yuan et al.²⁴⁵ demonstrated that the stereoselective C–C oxidative coupling of α -aryl ketonitriles can be accomplished over CsPbBr_3 perovskite QDs as a photocatalyst under exposure to visible light. In this study, to examine and compare various QDs in photocatalysis of the C–C oxidative coupling of α -aryl ketonitriles, eight types of QDs with varied compositions and capping ligands were developed. In particular, the zwitterionic ligand (that is, 3-(N,N-

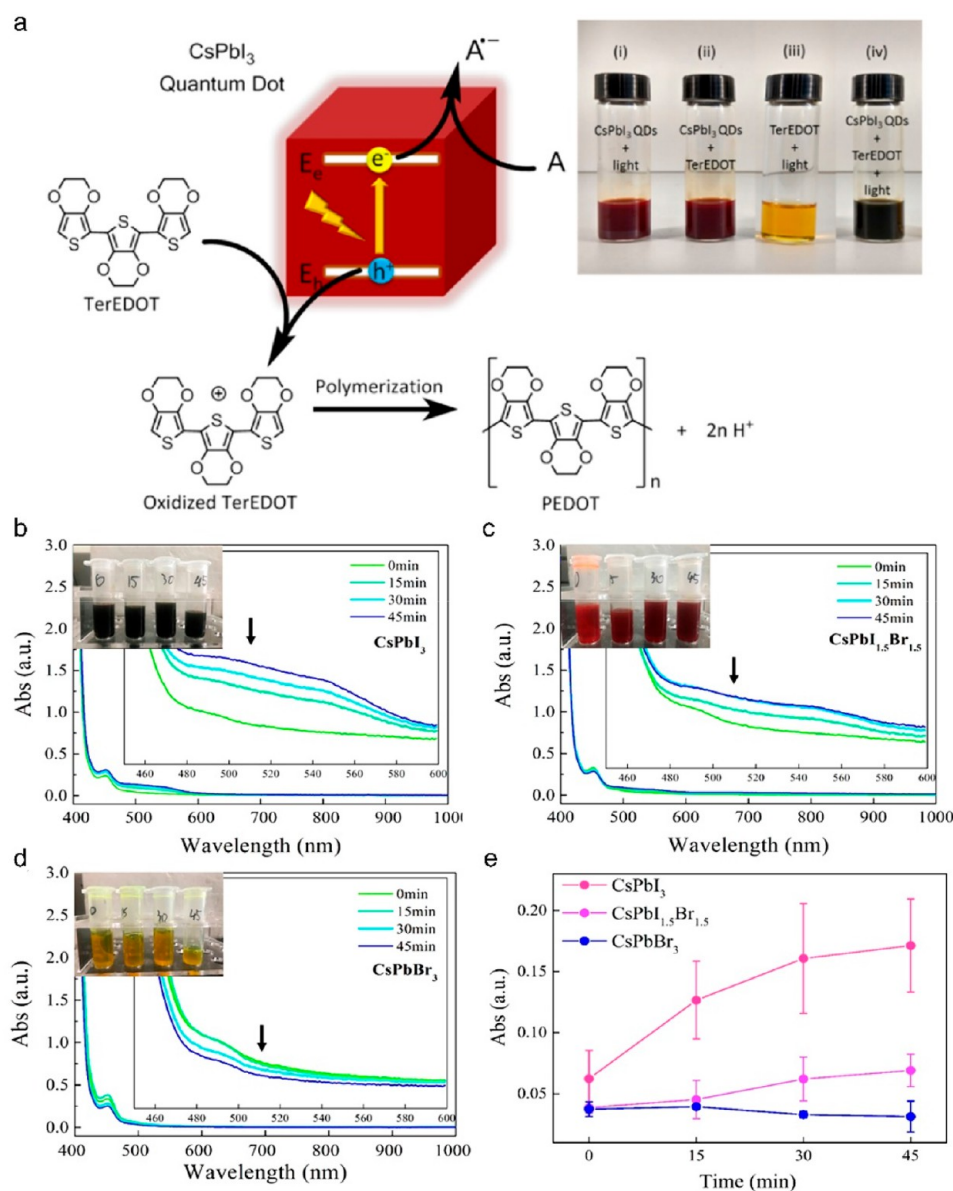


Figure 15. (a) Illustration of the proposed mechanism for photocatalytic polymerization of TerEDOT over CsPbI₃ QD under visible-light illumination. Reproduced with permission.²⁵⁷ Copyright 2017, The American Chemical Society. Time-dependent UV-vis spectra of the photocatalytic reaction of (b) CsPbI₃ p-QDs, (c) CsPbI_{1.5}Br_{1.5} p-QDs, (d) CsPbBr₃ p-QDs, and (e) the absorbance profile over reaction time at 510 nm for tracking PEDOT formation with different p-QDs. Reproduced with permission.²⁵⁸ Copyright 2019, The American Chemical Society. ^aA indicates the electron acceptor. E_g indicates the lowest energy electron state, and E_h indicates the highest energy hole state. Inset: photographs of (i) CsPbI₃ QD under light illumination for 90 min; (ii) CsPbI₃ QD and TerEDOT in the absence of light; (iii) the solution of TerEDOT after light illumination for 90 min; and (iv) CsPbI₃ QD and TerEDOT after light illumination for 90 min. The solvent is a mixture of dichloromethane and toluene (v/v, 1:1).

dimethyloctadecylammonio)propanesulfonate, DMOA-PS) capped colloidal CsPbBr₃ PQDs (ZW-CsPbBr₃) were achieved manipulating an already documented report.²⁴⁶

They discovered that the surface tuning of CsPbBr₃ PQDs with zwitterionic ligands can lead to the simultaneous improvement of the NC stability and rate of coupling reaction as well as the stereoselectivity. Moreover, a mechanistic insight study confirmed that for effective dimerization electron donors or large conjugated π systems were essential parameters. The study claimed that the reaction experiences a radical intermediate reaction route on the surface of the PQDs, implying that a less steric hindrance brings forth the desired *dl*-isomer. The constructed dimers, which are seldom accessible

using other oxidants or catalysts^{247,248} can be utilized as intermediates to form indenenes with significant pharmaceutical potentials.²⁴⁹ This study revealed that colloidal PQDs with suitable alterations can be favorable photocatalysts in organic manufacturing processes.²⁴⁵

3.2.5. Photocatalytic Polymerization of 3,4-Ethylenedioxythiophene. The synthesis procedure of halide PQDs necessitates the presence of surfactants to not only boost their durability but also preserve their morphology.^{250,251} Despite that, the surfactants may interfere with or even obstruct the interfacial charge carrier flow. Encapsulation is an effective tool to preclude the aggregation of the PQDs. The two frequently employed materials for the stabilization of

PQDs are insulating organic^{252,253} and silica protecting layers,^{254–256} both of which can hamper the flow of charge carriers between PQDs and the other functional materials, which means restricting their application. To resolve this issue, wrapping the QDs within a conductive protecting layer could be an alternative strategy. It can be expected that the protecting layer simultaneously encourages electron transfer, maintains the morphology, and enhances the durability of the PQDs.

Poly(3,4-ethylenedioxythiophene) (PEDOT), a commonly utilized conducting polymer in optoelectronic devices, is generally fabricated via electro/chemical oxidation of 3,4-ethylenedioxythiophene. Nonetheless, these techniques are not suitable for incorporating the PQD into the PEDOT structure; in that, the oxidizing environment may damage or even devastate the PQD. Hence, by virtue of the long-run durability of carriers and decent absorption coefficient of halide perovskites, the photodeposition route seems a rational solution to introduce the PQD into the structure of PEDOT. Further, it is postulated that photodeposition takes the advantage of photoinduced electron transmission from the PQD to the polymer, boosting the charge carrier transfer and thereby giving forth enhanced optoelectronic properties.²⁵⁷

As stated by Chen and colleagues²⁵⁷ all-inorganic halide perovskites with higher chemical stability as compared to organic–inorganic hybrid halide perovskites but identical preparation procedures are considered more favorable photocatalysts. In this regard, they examined both the photocatalytic activity and phase stability of all-inorganic perovskite for the polymerization of 2,2',5',2''-ter-3,4-ethylenedioxythiophene (TerEDOT) to PEDOT, which is usually manipulated as a hole transfer layer for manufacturing of the PSC. The photocatalytic process is presented in Figure 15a. For the photocatalytic reaction, TerEDOT and 1,4-benzoquinone (Qu) or molecular oxygen were used as the hole and electron acceptors, respectively.

Initially, the photocatalytic reaction was conducted in the open air in the presence of CsPbI₃ QDs. The findings showed that the increase in absorbance can be even more noticeable once both molecular oxygen and the CsPbI₃ QDs are present. This outcome substantiated that the synergy between molecular oxygen and CsPbI₃ QDs can indeed improve the polymerization process. The authors correlated this phenomenon with the formation of O₂^{•−}. For this, it was found that dissolved oxygen/Qu can function as an electron acceptor and accepts one electron to generate O₂^{•−}²⁵⁹ with which the oxidization of TerEDOT can take place and the polymerization can be expedited.

The photoinduced electron transfer from the TerEDOT precursor to the CsPbI₃ QDs plays an important role in the reaction process. It is well-known that the charge carrier transfer across the interface between different functional layers of the optoelectronic device is critical for the device's performance.²⁶⁰ The photoinduced electron transfer may optimize exciton dissociation in hybrid interfaces in devices^{261,262} and thus improve the interfacial carrier transfer. Thus, the presented photocatalytic reaction displays potential applications in optoelectronic material preparation and device fabrication.

The principal drawback of the common long-chain OA/OA ligands is their poor conductivity, which could slow down the charge flow between the QDs and hole/electron acceptor.^{263,264} Hence, the logical treatment of the surface

ligands can encourage the charge flow and exciton separation, for which a simple strategy has been documented by treating the QDs with MeOAc.¹⁸⁵ As a result, the replacement of the OA ligand with a shorter acetic acid may accelerate the charge flow between QDs and hole/electron acceptor. To accelerate the polymerization process of TerEDOT, inorganic LHPQDs (pristine-QDs, p-QDs) with various halide compositions were constructed and employed as photocatalysts. The inorganic LHPQDs treated with MeOAc (MeOAc-QDs) proved an enhanced photocatalytic activity thanks to the exchange of the OA ligand with shorter acetic acid and/or the defects created in the post-treating step that expedite the charge flow and restrict the notorious recombination of electrons and holes.

Li et al.²⁵⁸ prepared a series of CsPbX₃ p-QDs (X = I, I_{0.67}Br_{0.33}, I_{0.5}Br_{0.5}, I_{0.33}Br_{0.67}, Br) based on previously documented work with some modifications.²³⁴ To explore the role of halide composition on the qualities of QDs, the morphology was regulated in uniform by adjusting the reaction time. The photocatalytic investigations were performed to ascertain the role of halide composition in the photocatalytic performance of the CsPbI₃, CsPbI_{1.5}Br_{1.5}, and CsPbBr₃ p-QDs for the polymerization of TerEDOT to PEDOT. It was found that CsPbI₃ had higher photocatalytic activity relative to CsPbI_{1.5}Br_{1.5} PQDs, whereas CsPbBr₃ p-QDs demonstrated extremely negligible performance (Figure 15b–e). This finding is in accord with the fact that the narrower the bandgap the higher the photocatalytic activity as a result of the broad absorbance window.

3.2.6. Photocatalytic NO Removal. Tremendous attempts have been made to design an effective and reliable method for NO removal. Catalytic elimination of NO contaminations is accepted as an encouraging scheme, in which some developments have already been obtained in a few years. The NO removal induced by photocatalytic oxidation using semiconductors under visible-light exposure is a promising approach; however, it still suffers from low conversion efficiency. The bandgap and the locations of the conduction band and valence band are the two critical attributes of the photocatalysts to make sure whether the coming photon energy can be put to work or not. The proper bandgap guarantees the efficacious visible-light absorption and the locations of the CB and VB redox competence of the photocatalysts.

Despite several photocatalytic-oriented advantages of hybrid PQDs, their utilization as photocatalysts for the direct degradation of NO has rarely been explored. In this regard, hybrid PQD-based photocatalysts have received enormous research interest. Not to mention that their insufficient stability toward air, temperature, and light exposure resulting from high surface energy and myriad unpaired bonds at the surface impedes their mass-scale implementation.²⁶⁵ Up until now, plenty of schemes such as surface modification engineering,^{183,256} core–shell structure,²⁵⁵ and the embedment in microcrystals or glass,^{266–268} have been designed to enhance the tolerance of HPQDs. The formamidinium lead halide (FAPbX₃, X = Cl, Br, I) a member of the large family of PQDs possesses the characteristic credits of methylammonium lead halide MAPbX₃ (MA = CH₃NH₂, X = Cl, Br, I),²⁶⁹ and at the same time ensures the tolerance to some extent. This can be explained by the fact that FA has a larger size compared to MA, meaning that ion exchange favors the composition containing FA cation, forming a more symmetric crystal structure and a broader absorption spectrum.²⁷⁰

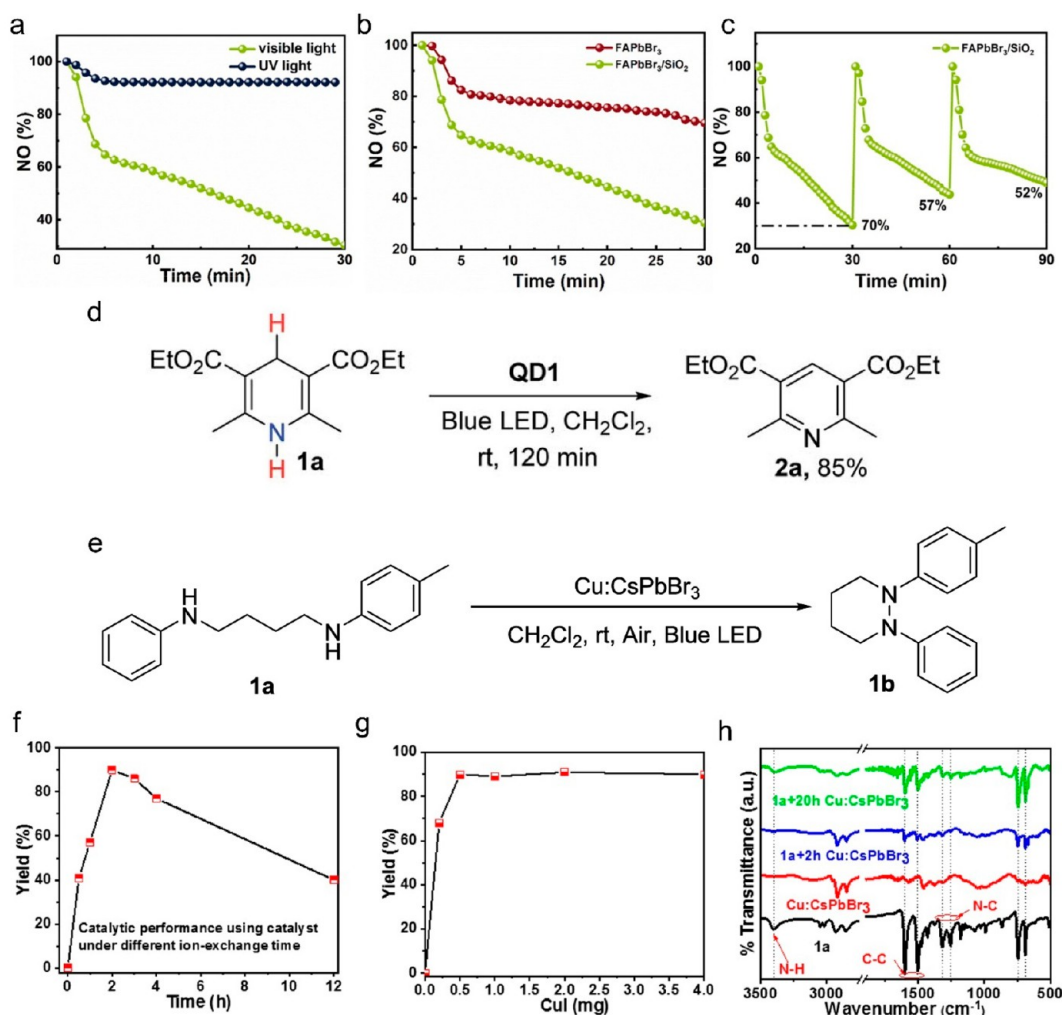


Figure 16. (a) Photocatalytic activity of FAPbBr₃/A-SiO₂ composites under visible light and UV light. (b) Photocatalytic performance of pristine FAPbBr₃ and FAPbBr₃/A-SiO₂ composites under visible light. (c) Stability test of FAPbBr₃/A-SiO₂ composites in three consecutive runs under visible light. Reproduced with permission.²⁶⁵ Copyright 2020, Elsevier. (d) Initial result of oxidative aromatization of **1a** in the presence of QD1 acting as the photocatalyst. Reproduced with permission.¹⁸⁴ Copyright 2021, The Royal Society of Chemistry. (e) Formation of pyridazine **1b** from surface-bound diamine substrates (e.g., **1a**). (f) Optimized Cu loading for yield of **1b**; 0.5 mg CuI per 2 mg NCs used for ion-exchange under different time. (g) Catalytic performance using excess CuI for ion-exchange with 2 mg NCs under 2 h ion-exchange time. (h) FTIR comparison indicating the surface binding of substrate through Cu. Reproduced with permission.²⁷¹ Copyright 2021, The American Chemical Society.

Zeng et al.²⁶⁵ recently conducted a one-pot synthesis route to attach QDs onto the amino-containing silica spheres (A-SiO₂), resulting in an improvement in the durability of random lasers built by the CsPbBr₃ QDs. Inspired by this, Huo et al.²⁶⁵ have also used the A-SiO₂ for a completely different story, yet a relatively similar application—that is—promoting the photocatalytic durability of the FAPbBr₃ in the NO degradation reaction. Therefore, manufacturing of the composite was carried out via attachment of FAPbBr₃ on A-SiO₂ via N–H bonding.

The photocatalytic performance of virgin FAPbBr₃ and FAPbBr₃/A-SiO₂ were assessed via varying the concentration of NO gas every 30 min under both visible- and UV-light irradiation. From Figure 16a, one can notice that under UV light, the material could deteriorate easily. In addition, the photogenerated electrons and holes recombine instead of assisting the charge carrier conduction. Figure 16b compares the photocatalytic capacity of neat FAPbBr₃ with that of FAPbBr₃/A-SiO₂, the presence of plentiful NH_x groups on the surface of A-SiO₂ alleviated the development of hybrid PQDs

nucleation, leading to less agglomeration of the bare FAPbBr₃ and in turn, availability of more active sites. In this way, the photocatalytic conversion performance is significantly boosted because of the N–H chemical bonding between NH_x and the FA cation. For this reason, the photocatalytic activity of FAPbBr₃/A-SiO₂ composites was much higher than their bare counterpart, FAPbBr₃ for the NO removal.²⁶⁵ Finally, Huo et al.²⁶⁵ attempted to examine the durability and recyclability of photocatalytic performance by multicycle experiments for FAPbBr₃/A-SiO₂ photocatalyst (Figure 16c). The NO removal ratios were calculated to be 70%, 60%, and 57% from the first to the third cycles, respectively.

3.2.7. Visible-Light Organic Transformations. In addition to all the above-mentioned applications, the LHPs can activate and catalyze the organic substrates in photoredox reactions in which light absorption, charge separation, and transfer are of high interest. Until now, studies conducted on organic transformation via photocatalysis have all exploited polydisperse LHP NCs (ca. 2–100 nm), which are not included in quantum confinement regimes, owing to their large sizes of

crystals.¹⁸⁴ However, by contrast with these large-sized PNCs, colloidal LHPQDs fabricated by common techniques have not shown decent catalytic efficiency, indicating their low performance against moisture, air, substrates, and solvent. There have been numerous efforts to stabilize CsPbI₃ and CsPbBr₃-based perovskites with the aid of some additives such as halide salt,²⁷² ammonium halide,²⁷³ 2,2'-iminodibenzoic acid,²³ phosphinic acid,¹⁷⁵ sulfides and metal ion,²⁷⁴ and polymers²⁷⁵ or adopting the application of specific postsynthetic purification steps.^{46,276} Despite the success of these approaches in SCs and LEDs, they are not practical candidates for photocatalysis.

Nonetheless is air a benign source of oxygen that can function as an oxidizing agent in a singlet state^{277–279} perovskite-based CQDs prepared by conventional methods, and bulk halide perovskites are extremely susceptible to air and moisture, impeding their widespread use in photocatalysis.^{143,280–282} Those photocatalytic reactions accompanying nitrogen heterocycles such as 1,2-dihydropyridines (1,2-DHP), 1,4-dihydropyridines (1,4-DHP), dihydrobenzothiazoles and pyrazolines are critical; nevertheless, they are extremely sluggish; meaning that either an effective photocatalyst or a strong oxidant should be implemented. Earlier studies' focus on the aromatization of 1,2-dihydropyridine (1,2-DHP), 1,4-DHP, and 1,3,5-triaryl-pyrazolines as substrates were to some extent successful^{283–286} using Ru-, Pd-, and Pt-based molecular catalysts but at the expense of high costs imposed by the noble metal catalyst that are not tolerant against oxygen, which rules out the possibility of recyclability. To this end, Pradhan et al.¹⁸⁴ reported the manufacturing of the air-tolerant CsPbBr₃ perovskite CQDs (QD1) via the "three precursors method" and their passivation with the help of halide to oxidize and subsequently aromatize a wide spectrum of azaheterocycles.

To appraise the photocatalytic value of QD1 in aromatization reactions under air exposure, the authors selected Hantzsch ester, i.e., diethyl 2,6-dimethyl-1,4-dihydropyridine-3,5-dicarboxylate (Figure 16d, 1a) as a model substrate on account of its synthetic accessibility. The production of 2a with 85% yield was verified by nuclear magnetic resonance (NMR) (Figure 16d). To further shed light on the importance of the presence of a catalyst for the aromatization of 1a (Figure 16d), the reaction was conducted in the absence of the catalyst as well. The results demonstrated only a 20% yield of 2a production which pointed out the crucial role of QD1 to catalyze the reaction. In addition, the reaction was carried out in the absence of light and a negligible amount of 2a was observed, implying the significance of light for the completion of the reaction.

For another important organic reaction, Martin et al.²⁷¹ developed a novel heterogeneous photocatalytic system that exploited a lead-halide perovskite NC to absorb photons and encouraged the transfer of the photogenerated holes to a surface-bound transition metal Cu-site, bringing about an N–N heterocyclization reaction. In this work, the authors employed a controlled cation exchange reaction to fabricate a unique "merged" photocatalysis that utilized the NCs as the light absorber decorated with Cu catalyst incorporated onto the NC surfaces. The photocatalytic NCs were fabricated by exposing as-synthesized CsPbBr₃ NCs to a solution of Cu(I)X (X = Cl, Br, I) at ambient temperature, resulting in lightly doped Cu:CsPbBr₃ NCs.

An intramolecular N–N heterocyclization was examined to inspect the catalytic activity of the Cu:CsPbBr₃ NC system.

Such reactions can be directly rendered from surface-bound diamine substrates (e.g., 1a, Figure 16e), in the presence of ambient air and visible light as the terminal oxidant and the energy source, respectively. The photocatalysis showed that the Cu:CsPbBr₃ photocatalyst suspended in dichloromethane under air with 450 nm LED irradiation, without any other additives, produced pyridazine 1b from 1a (Figure 16e) with 90% yield throughout 18 h. the control experiments excluding any component of the catalytic system, light, or air did not give rise to any product. From the combination of various reaction conditions, overall, a 2 h CuI-exchange brought about the highest yield of 1b (Figure 16f). The authors found that a higher Cu-loading did not necessarily correspond to a higher reactivity. The optimal catalytic activity was attributed to a balanced and synergistic effect of the Cu cation exchange and Cu catalytic activity. Imperfect Cu-loading indicated a lower number of Cu-catalytic sites and accordingly a lower yield. On the contrary, higher Cu incorporation gave rise to phase segregation, forming CsCuX₃ rather than further substitution of Cu in the CsPbBr₃.

With the designed ion-exchange system, the amount of Cu-incorporation was determined via the exchange reaction time, that is, when an excess of Cu(I) was applied but under the same exchange, the same photocatalytic activity (yield of 1b, Figure 16g) was found. The observations showed that only cation-exchanged Cu sites were responsible for the catalytic reaction and one that stood out among other samples, appeared to be Cu_{0.01}:CsPbBr₃. The origin of oxidative intramolecular N–N coupling is the diradical formation that must take place on a single diamine molecule. To achieve this, it was suggested that the diamine substrate coordinates to Cu-sites on the surface of the perovskite NCs in such a manner that consecutive excitation of the NC leads to the diradical formation. The diamine coordination to Cu surface sites was directly proved by an IR analysis, the comparison between pristine CsPbBr₃ and Cu-exchange NCs (Figure 16h).

3.2.8. Stability Issues of PQDs under Different Environments. Considering that the stability of PQDs under different environments in photocatalysis plays a pivotal role, hereunder the protecting methods to modify and engineer the APbX₃ PQDs are addressed. Passivation of ligands or modifying ligands to protect APbX₃ PQDs from degradation is normally carried out by ligand exchange, which is a well-known approach for this purpose. Ligands used for the preparation of PQDs have a strong effect on the stability and dispersibility of PQDs in various solvent media.²⁸⁷ Moreover, long chain ligands like OA and OLA, which are the prominent capping ligands for the synthesis of CsPbBr₃ PQDs, are highly vulnerable to exposed air, water, light, and temperature.^{288,289} To this end, ligand exchange is an established method among chalcogenide quantum dots.^{290,291} Recently, people have adapted and extended this strategy to PQDs with the utilization of different types of ligands.^{292,293}

Types of ligands based on the different properties and applications required could be as follows, short chain-monofunctional, bidentate, bifunctional, multidentate, and so on. Ligands with high molecular length can form larger interparticle spacing inducing lower charge/energy transfer.²⁹⁴ Hence, employing short chain alkyl ligands²⁹⁵ is desired for photo applications. Ligand exchange with short-chain ligands like decanoic acid (DA),²⁹⁶ cinnamic acid,²⁹⁷ 2-hexyldecanoic acid, and phosphoric acid²⁹² provide facilitated charge transfer, lower recombination, and longer carrier lifetimes. Nonetheless,

they can only enhance the stability by a couple of hours, which cannot be applied to various solvent systems. Ligands that contain bicarboxylic acid groups namely succinic acid (SA)²⁹⁸ have been found to exhibit enhanced stability and optical properties.²³

From the literature, bidentate ligands have shown both core stability and ligand stability. For instance, zwitterionic ligands like alpha amino butyric acid, iminodibenzoic acid,^{23,246} and lysine allow both functional groups in them to bind to Pb²⁺ ions, enhancing fluorescence and stability.²⁹⁸ In 2020, the Zhang group²⁹⁹ documented air-stable CsPbBr₃ PQDs by introducing an L-lysine ligand via a spin coating method. The resulting material showed that L-lysine had stronger binding energy to CsPbBr₃ PQDs than OA binding. This resulted in reduced recombination and enhanced the carrier lifetime. The surface passivated L-lysine CsPbBr₃ PQDs material demonstrated improved environmental stability for more than 45 days (in exposed air with high humidity) and 500 h under the illumination of daylight.

In another work, perfluoro decanoic acid (PFDA) and DA ligands were exploited to promote thermal stability and strengthen the tolerance of CsPbBr₃ PQDs against polar environments.²⁹⁶ The fluorocarbon component easily coordinates with metal ions, causing the reduction of surface defects. Their thermal stability study also showed good enhancement. PFDA-CsPbBr₃ PQDs were subjected to 100 °C for 4 h. The PL intensity was constant at 92.1% for PFDA-CsPbBr₃ PQDs, while it showed a drastic reduction for pristine CsPbBr₃ PQDs and DA-CsPbBr₃ PQDs. Resistivity in polar solvents (ethanol) followed the same trend with PL maintained at 36.5% for PFDA- CsPbBr₃ PQDs while it reduced to 9.53% and 13.6% for the other two. Thus, surface passivation using a PFDA ligand conferred good photo and thermal stability together with solvent resistivity, helping in the construction of stable electroluminescent devices.

In addition to the above-written research, Pan et al.³⁰⁰ introduced a quite durable CsPbX₃ QD film with the aid of a halide ion pair (such as didodecyl dimethylammonium bromide (DDAB)), which was a relatively short ligand. These types of ligands are effective for charge transport in the QD film which can be good candidates for the development of highly efficient perovskite LEDs. This group found out that surface-treated CsPbI₃ with 2,2'-iminodibenzoic acid (IDA) contains a double carboxyl group compared to surface-capped CsPbI₃ with a single carboxyl group which means it was bound to two Pb atoms on the surface. The theoretical calculations validated that the double carboxyl-terminated surface has higher binding energy (1.4 eV) compared to the single carboxyl-terminated surface using an OA ligand (1.14 eV).²³

The structural stability of perovskite materials can be related to the Goldschmidt tolerance factor, which was mentioned earlier. From the reports, perovskites can preserve their 3D structure while the tolerance factor would be between 0.813 and 1.107.³⁰¹ The change from one crystal structure to another can be pertained to this factor and with slight structure distortion. However, slight distortion may not be sufficient to modify the photoelectric properties of the material's energy band. In this sense, a dramatic change in the A-site of ABX₃ could result in the collapse of the perovskite structure.³⁰² This phenomenon is even more evident when it comes to lead iodine perovskite materials (CsPbI₃, MAPbI₃, and FAPbI₃). The 3D lead iodine perovskite materials are thermodynamically

metastable at room temperature, easily transformed into the 1D structure of a broad band, and this incident can be escalated as the crystal size shrinks.^{270,303} Similar to studies reported on CsPbBr₃ PQDs, ligand modification has a strong impact on the stability of other APbX₃ PQDs. For instance, Tan et al.³⁰⁴ developed the CsPbX₃ PQDs using octylphosphonic acid (OPA) rather than conventional OA and OLA ligands. Their results showed that OPA-CsPbX₃ quantum dots can not only achieve high-quality dispersion in solvents after several purification treatments but also exhibit ultrahigh PLQY (>90%) because of the powerfulness of OPA and lead ions effect.

In addition, Jasieniak et al.¹⁷⁵ used bis(2,2,4-trimethylpentyl) phosphinic acid (TMPPA) for the preparation of CsPbI₃ PQDs instead of an OA ligand. They found that the PL intensity of CsPbI₃-TMPPA sustained its initial value after 20 days compared to the decomposition of CsPbI₃-OA after 3 days. It was shown that the following molecules with large steric hindrance can induce high proton solvent resistance in PQDs; highly branched (3-aminopropyl) triethoxysilane (APTES),³⁰⁵ polyhedral silsesquioxane-[3-(2-aminoethyl) amino] propylheptylbutyl substitution (NH₂-POSS),³⁰⁶ 1-tetradecylphosphonic acid (TDPA),³⁰⁷ 2-adamantyl ammonium bromide (ADBr),³⁰⁸ trioctylphosphine oxide (TOPO),³⁰⁹ mercapto- β -cyclodextrin (SH- β -CD),³¹⁰ poly(lactic acid) (PLA),³¹¹ and cage polyhedral oligosilsesquioxane (POSS).²⁵⁶ The experiments verified that highly branched ligands can obstruct the penetration of proton solvent into the surface of PQDs. As a whole, ligand exchange is an effective and simple scheme for the surface passivation of PQDs to fine-tune surface compatibility, creating PQDs with improved properties to stand in air, water, light, and heat. It must be noted that the binding energy of the new ligand with the core ought to be much stronger than the existing system.

Apart from this, an aqueous solution is the main medium for photocatalysis. However, LHPQDs are poor materials in contact with water. To overcome this challenge, Zhang et al.³¹² came up with a brilliant idea, which was to embed CsPbX₃ (X = Cl, Br, I) NCs in polystyrene-based microhemispheres (MHSs) to produce "waterproof" NCs@MHSs hybrids. They initially used polyvinylpyrrolidone (PVP) to adsorb and wrap around the surface of PQDs to act as a protective layer. The PVP could enhance the quantum yield of CsPbX₃ NCs and at the same time acted as the interface layer to combine PQDs with polystyrene polymers. The prepared CsPbX₃ NCs@MHSs was demonstrated to be stable and nontoxic as multicolored luminescent probes in living cells. Finally, Yang et al.³¹³ wrapped CsPbX₃ (X = Cl, Br, I) NCs inside the phospholipids to acquire CsPbX₃-phospholipid micelles (CsPbX₃@phospholipids). The phospholipid layer endows CsPbX₃ NC with remarkable water resistance and the capability to further biofunctionalized and eventually be more biocompatible. Prepared CsPbX₃@phospholipid microclusters displayed strong fluorescence and narrow half-peak widths in water for over 4 months.

3.2.9. Summary. Halide PQDs have recently emerged as competent photocatalysts in the energy and environment-oriented fields including hydrogen production, CO₂ reduction, and organic compounds degradation to name but three following their exceptional properties such as superior light-absorbing factors and small binding energies of the exciton. Knowing that environmental concerns namely rapid depletion of fossil fuels, excessive emission of CO₂, and contamination

containing sewages released from industrial plants demand urgent solutions, e.g., photocatalysis by state-of-the-art PQDs under visible light to take the most advantage of renewable solar energy. Furthermore, PQDs can catalyze the organic substrates in photoredox reactions in which light absorption plays a pivotal role. Halide PQDs, regardless of their prominent traits, do not perform well under open air and in the vicinity of a wet environment. One approach that has been suggested is wrapping the QDs within a conductive protecting layer which not only can encourage electron transfer, but also maintains the morphology and enhances the stability of the PQDs. Moreover, the swift recombination of electrons and holes is another barrier to their far-reaching implementation. The solution to this impediment, i.e., the presence of long-chain OA/OLA that could slow down the charge transfer, has been responded to by the exchange of long-chain ligands with short-chain ligands to accelerate the charge flow between QDs and the hole/electron acceptor.

3.2.10. Future Perspectives. Despite striking properties coming along with the utilization of HPQDs, their practical photocatalytic applications require further investigations. The utmost challenging parameter in PQDs' widespread utilization is their poor stability in the air or under aqueous solutions. To overcome this impediment, efforts have been made. For this, surface modification engineering, a core-shell structure, and the embedment in microcrystals or glass are proposed schemes that have been designed to enhance the tolerance of HPQDs. Reports revealed that coupling LHPQDs with graphene oxide or $g\text{-C}_3\text{N}_4$ brings forth efficient charge separation, leading to an enhanced catalytic activity toward CO_2 reduction. Moreover, the surface tuning of CsPbBr_3 PQDs with zwitterionic ligands can lead to the simultaneous improvement of the NC stability and rate of coupling reaction as well as stereoselectivity. Thus, the addition of a second component to HPQDs or surface and compositional engineering can alter the structural and chemical features, resulting in a more stable PQD-based photocatalyst.

Even though Pb-based HPQDs have been the archetype in PQD-derived photocatalysts, the wide applications of halide PQD-based materials are restrained due partially to the presence of the Pb component. Substitution of Pb with other environmentally benign elements can overcome this bottleneck and pave the way for nontoxic PQD-based photoelectrochemical devices. For this, the combination of various photophysical characterizations and DFT calculations to deepen our insights into the basic luminescence properties and dynamic processes of halide PQDs can be helpful for the fabrication of high-quality materials. The proposed Pb-free PQD photocatalyst not only should hold a competent photocatalytic capacity, but also should bear sufficient stability under different reaction environments.

The real application of photocatalysts based on HPQDs is subject to favorable photocatalytic attributes such as efficient light absorption, charge generation, separation, and transfer as well as durability. Encouraging one aspect may generate other challenges. For instance, the synthesis procedure of HPQDs necessitates the presence of surfactants to not only boost their durability but also preserve their morphology. Despite that, the surfactants may interfere with or even obstruct the interfacial charge carrier flow. Encapsulation is an effective tool to preclude the aggregation of the PQDs. To resolve this issue, wrapping the QD within a conductive protecting layer could be an alternative strategy. It can be expected that the protecting layer simultaneously encourages electron transfer, maintains

the morphology, and enhances the durability of the PQDs. Based on the discussion mentioned above, material/compositional engineering should be at the forefront.

4. CONCLUSION

In summary, even though there has been an explosion of research on the development of next-generation SCs, and photocatalysts based on halide PQDs in the span of a few years, their real application is still a distant dream due to many key challenges that require further consideration. With the progress of the research, the photo-, solvent-, and thermal-tolerance of halide PQDs still lag behind traditional QDs, which demand to be addressed immediately and properly. Thanks to various studies in the PQDSCs up to now, the enhanced electron coupling, charge transport, and reduced recombination loss in QD film led to improved device performance. With the tremendous efforts in the perovskite optoelectronic device research, the PQD-based SCs, LEDs, lasers, and photodetectors have mainly been the center of attention, while they can be realized in top-named photocatalytic reactions such as hydrogen and oxygen evolution. Taking into account the mentioned pros and cons, halide PQDs have opened a new era, especially in the fields of SCs and photocatalysis. Thereupon, despite persistent flaws, brilliant solutions are already underway.

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Notes

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